SEC P vs NP log 2026-04-27

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 23:46 UTC

SEC P vs NP — daily log 2026-04-27

55 cycles recorded

Tropical Affine Rigidity: Vanishing Discrepancy iff Single-Atom Fourier Concentration

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Fourier Analysis (Tropical Geometry / Idempotent (Maslov) Harmonic Analysis) × Affine-Function Recognition Decision Problem (membership in the AC^0-decidable class of degree-1 tropical polynomials, equivalently the zero-discrepancy promise problem)
- **Recorded**: 2026-04-27 00:22 UTC
- **Entry ID**: `81928f9c440e`

Statement

Let f be a TropicalPolynomial over a finite domain D = {-N, ..., N} ⊂ Z (min-plus semiring). Let F = FourierTransform(f) denote its TropicalFourierTransform (Legendre–Fenchel transform: F(y) = min_{x in D}(f(x) - x*y) over a finite slope grid Y), and let m(f) = MinimalFourierCoefficient(f) = min_{y in Y} F(y). Then DiscrepancyCalculation(f) = 0 if and only if f is tropically affine, i.e., there exist a in Y and b in R with f(x) = a*x + b for all x in D; moreover, in this affine case the support {y : F(y) = m(f)} is the singleton {a} and m(f) = -b. Equivalently, DiscrepancyMeasure(f) > 0 iff the level set argmin_y F(y) has cardinality unequal to 1 in the regular position OR F attains its minimum -b at exactly the unique slope a only when f is affine — formally: Discrepancy(f) = 0 ⇔ |{y in Y : F(y) = m(f)}| = 1 ∧ f(x) ≡ (argmin_y F(y))*x − m(f).

Rationale

This sub-conjecture sharpens axiom A3 (DiscrepancyMeasure bounded by max |Fourier coefficient|) at its extremal saturation point. A3 says discrepancy is controlled from above by Fourier magnitudes; the natural converse extremal case is when discrepancy collapses to zero: this should force the Legendre–Fenchel transform to be a singular Dirac-like object on the slope grid, since affine functions are the unique 'tropically uniform' (curvature-free) elements. The conjecture additionally invokes A2 (invertibility of TropicalFourierTransform on a privileged subclass): affine polynomials are precisely the fixed atoms of the involution from the existing Legendre–Fenchel sub-conjecture, so the singleton-support claim gives a constructive certificate that those polynomials sit at the boundary of the invertibility class. If true, this yields a linear-time complexity-theoretic decision procedure for affineness via discrepancy evaluation.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1207.2443v2] Tropical Teichmuller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [2006.04823v3] Quantum Legendre-Fenchel Transform - [0710.0377v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume I) - [0709.4119v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume II)

Empirical Test

- exit code: `0`, elapsed: `5.79s`


```
"instances_tested": 600, "conjecture_holds": true, "counterexample": ""}]}
{"seed": 23, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 37, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 53, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 71, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
RESULT: INCONCLUSIVE not enough seeds supported
```


Judge reasoning

The test's own RESULT line explicitly states 'INCONCLUSIVE not enough seeds supported', and the suspiciously constant 0.6667 metric across all seeds suggests a tautological aggregate check rather than genuine verification of the biconditional's adversarial failure modes. | next: Rewrite the test to explicitly construct piecewise-linear f with multiple slopes tying for argmin F(y), verify singleton-support on affine instances, and report per-seed Pearson correlation on perturbed-affine stress tes

---

Continued Fraction Depth of Spectral Ratio Bounds Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine approximation (continued-fraction expansions and partial-quotient growth, in the spirit of Khinchin and Gauss-Kuzmin metric theory) × Discrepancy-based randomized two-party communication complexity lower bounds for ±1 sign matrices
- **Recorded**: 2026-04-27 00:31 UTC
- **Entry ID**: `71ce01b16c1e`

Statement

Let M ∈ {-1,+1}^{N×N} with N=2^n have singular values σ_1 ≥ σ_2 ≥ … ≥ σ_N, and let ρ(M)=σ_2/σ_1 ∈ [0,1]. Let (a_1,a_2,…,a_K) be the simple continued-fraction expansion of ρ(M) truncated at K=⌈log_2 N⌉ partial quotients (computed via the Euclidean algorithm on a 4N-digit decimal truncation of ρ), and define S(M)=Σ_{i=1}^{K} log(1+a_i). Then for every n≥3 and every ±1 sign matrix M of size $2^{n\times 2}$n, disc(M)·N ≥ c · σ_1(M) / (1 + S(M)) for an absolute constant c=0.05, where disc(M) is the combinatorial rectangle discrepancy of M.

Rationale

Highly structured matrices (Hadamard, Sylvester, AND/XOR-lifted indicators) have spectra that are integer or low-height rational, forcing σ_2/σ_1 to be exactly 0, 1, or a fraction with tiny partial quotients — yielding small S(M) and hence the well-known large disc·N ≈ σ_1. Generic matrices satisfy Gauss-Kuzmin statistics with partial quotients distributed like 1/(k(k+1)) and S(M)=Θ(log N), exactly compensating their σ_1=Θ(√N) into disc·N=Θ(√N). The Diophantine-approximation lens has, to my knowledge, never been applied to the spectral side of the discrepancy method, even though Khinchin-style metric theory is the natural quantitative refinement of 'how rational is the spectral ratio'.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1210.1461v1] A Scalable CUR Matrix Decomposition Algorithm: Lower Time Complexity and Tighter Bound - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2209.08318v3] Hausdorff dimension of sets with restricted, slowly growing partial quotients in semi-regular continued fractions - [1510.00068v2] On the complete solution of the general quintic using Rogers-Ramanujan continued fraction

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 137, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 113, in run_trial
    M = M_type()
        ^^^^^^^^
TypeError: run_trial.<locals>.<lambda>() missing 1 required positional argument: '_'
```

Judge reasoning

The test crashed with a TypeError (lambda missing positional argument '_') before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the M_type lambda signature in run_trial so all four ensembles can be instantiated uniformly, then rerun the 140-triple sweep.

Young-Flattening Rank Gap Lower-Bounds Padded-Permanent Border Rank

- **Verdict**: INCONCLUSIVE
- **Bridge**: Geometric Complexity Theory via Young flattenings (Landsberg-Ottaviani equivariant rank obstructions) × Border rank of the padded m x m permanent tensor Per_m viewed as a slice of det_n
- **Recorded**: 2026-04-27 01:05 UTC
- **Entry ID**: 0e80f2d2c50a

Statement

Let Per_m be the m x m permanent and det_n its n x n padded determinant embedding (n >= m^2). Let YF_k(T) denote the rank of the Young flattening of tensor T indexed by the partition (k, 1^{m-k}) acting on the symmetric m-th power of C$^{\{m}$2}. Conjecture: for all 2 <= m <= 6 and all 1 <= k <= m-1, the equivariant rank gap delta(m,k) := YF_k(det_n viewed as Per_m-padding) - YF_k(Per_m) satisfies delta(m,k) >= c * binomial(m,k) * (n - m^2 + 1) with absolute constant c = 1/2, and consequently the border rank lower bound brk(Per_m) >= max_k YF_k(Per_m) / binomial(m-1, k-1) is tight to within a factor of m.

Rationale

Young flattenings are the only currently known equivariant obstructions that have produced super-additive border-rank lower bounds for permanents (Landsberg-Michalek), yet no quantitative law connects their rank gaps to the padding parameter n; pinning down delta(m,k) would convert orbit-closure containment questions into a finite linear-algebraic check. Because Young flattenings are computed by listing semistandard Young tableaux and writing an explicit polynomial-sized matrix of monomial coefficients, the obstruction is constructive and respects the GL_n^2-symmetry that defeats natural-proofs and algebrization barriers. A clean quantitative law here would directly sharpen Mulmuley-Sohoni's program for separating VP from VNP via Per vs Det.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [1993636.1993704] Geometric complexity theory and tensor rank - [s2:1011.1350] Geometric Complexity Theory and Tensor Rank

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 50, in run_trial
    YF_k_det_n = young_flattening_rank(det_n, m, 1)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 39, in young_flatten
    return len(gaussian_elimination(M))
               ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 24, in gaussian_elim
    factor = -A[j][i] / A[i][i]
              ~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before producing any delta(m,k,n) data, so the pre-registered support condition could not be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the gaussian_elimination routine (handle zero-pivot/short-row cases, e.g., add partial pivoting and bounds checks) and rerun the deterministic sweep over m∈{2,3,4}, 1≤k≤m-1, n∈{m²,...,m²+4}.

Pebble-Cost Gadget Lifts Decision Tree Depth to KW-Game Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Black-white pebbling games on rooted DAGs (Cook-Sethi pebbling cost) × Karchmer-Wigderson communication complexity on lifted query problems
- **Recorded**: 2026-04-27 01:38 UTC
- **Entry ID**: `52cebd2ffe62`

Statement

Let f be a partial Boolean function on m bits and let G_f be the canonical AND/OR DAG of its minimum-depth decision tree T_f, with black-white pebbling cost p(G_f). Define the lifted relation f∘g where g is the 2-bit indexing gadget IND_2 on b=2 address bits, and let KW(f∘g) denote the deterministic KW-game length on f∘g. Then for every f with |T_f| ≤ 2^m and m ≤ 12, KW(f∘g) =

depth(T_f) + p(G_f) ± 1, i.e. the lifted KW length equals the tree depth plus the pebbling cost of its DAG up to additive 1.

Rationale

Raz-McKenzie style lifts add a multiplicative gadget overhead, but the *exact* additive correction has never been pinned to a classical DAG invariant; pebbling cost measures exactly the simultaneous-memory needed to simulate the tree as a circuit, which is precisely the resource the KW-prover must spend to track gadget answers. If correct, this gives a tight (not asymptotic) lifting identity computable on tiny instances, sharper than known O(depth · log b) bounds. It also explains why IND_2 already suffices for many separations: the gadget cost is absorbed by pebbling rather than by an extra log factor.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 92, in run_trial
    m = random.randint(3, 5) if m == 6 else m
                                ^
UnboundLocalError: cannot access local variable 'm' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so the pre-registered support condition (≥200 samples with |k−d−p|≤1 for 100%) was never evaluated. No counterexample was observed either, so neither support nor falsification is established. | next: Fix the variable-scoping bug in run_trial (initialize m before the conditional reassignment) and rerun the full sweep over m∈{3,...,12} to obtain the required ≥200 samples.

Mealy Automaticity Lower-Bounds Truth-Table Formula Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite-state transducer minimization / 2-automaticity of binary sequences (Allouche–Shallit automaticity over the alphabet {0,1}, distinct from any compression-based complexity) × Meta-complexity: minimum {AND,OR,NOT}-formula leaf count L(f) of a truth table, used as a tractable MCSP proxy
- **Recorded**: 2026-04-27 02:14 UTC
- **Entry ID**: `33ac9cdf1529`

Statement

For an n-variable Boolean function f with truth table T_f ∈ $\{0,1\}^{\{2}$n}, let A(f) be the number of states of the minimum complete DFA over alphabet {0,1} that accepts L_f = { bin_n(k) : T_f[k] = 1 }, where bin_n(k) is the n-bit MSB-first binary expansion of k. Let L(f) be the leaf-count of the smallest {AND,OR,NOT}-formula computing f over the literals x_1,...,x_n,¬x_1,...,¬x_n. Conjecture:

for every n ≥ 2 and every f, A(f) ≤ (n+2)·L(f) + 2; equivalently, log_2 A(f) ≤ log_2 L(f) + log_2(n+2) + o(1). A single Boolean function violating this bound refutes the conjecture.

Rationale

Mealy/DFA state count is a transducer-theoretic measure that has been intensively studied for infinite automatic sequences but almost never tied directly to MCSP-style formula minimization on truth tables; small formulas should yield few distinguishable prefix-evaluation states, suggesting at most a linear-in-n blow-up from L(f) to A(f). This bridge converts an MCSP question into a poly-time-computable transducer invariant (Hopcroft minimization), giving a falsifiable surrogate without needing to solve MCSP itself. If the bound fails, the offending function exposes a 'state-rich, formula-cheap' object — a new natural property candidate that would not relativize, fitting the META_COMPLEXITY safe-direction profile.

Novelty

- Judge: `NOVEL` over 10 arXiv hits

Top hits consulted: - [2003.09703v1] Variance function of boolean additive convolution - [1112.2313v2] QBF-Based Boolean Function Bi-Decomposition - [2402.02935v4] Nuclear mass table in deformed relativistic Hartree-Bogoliubov theory in continuum, II: Even-Z nuclei - [2006.04708v2] On three conjectures of P. Barry - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 118, in run_trial
   A_f = hopcroft_minimization(set(range(2**n)), truth_table_to_dfa(truth_table, n)[1], set())[0]
                                                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 95, in truth_table_t
    transitions[s]['0'] = get_next_state(s, '0')
                          ^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 89, in get_next_stat
    for i in range(len(state)):
                   ^^^^^^^^^^

TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed with a TypeError in truth_table_to_dfa before any function was evaluated, so no A(f) or L(f) data was produced and the pre-registered support condition cannot be checked. | next: Fix the DFA construction (state representation type bug in get_next_state) and rerun the enumeration/sampling protocol over n∈{2,...,8}.

Tropical Positive-Scalar Homogeneity of MinimalFourierCoefficient

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: TROPICAL_FOURIER_ANALYSIS (tropical geometry × Fourier analysis over the max-plus semiring) × Min-plus circuit complexity / tropical circuit lower bounds (a complexity-theoretic measure governing shortest-path and dynamic-programming circuits)
- **Recorded**: 2026-04-27 02:28 UTC
- **Entry ID**: `6be01f5e483c`

Statement

Let f be a TropicalPolynomial on a finite grid $X \subset R^d$, and let $\lambda > 0$. Define the tropically rescaled polynomial $g = \lambda \odot f$, i.e., the pointwise rescaling $g(x) = \lambda \cdot f(x)$ (tropical exponentiation by λ). Let F = TropicalFourierTransform(f) and G = TropicalFourierTransform(g) be their tropical Fourier coefficients (the Legendre–Fenchel-type transforms guaranteed invertible by axiom A2 on the relevant class). Then MinimalFourierCoefficient(g) = λ · MinimalFourierCoefficient(f), and DiscrepancyCalculation(g) = λ · DiscrepancyCalculation(f). In particular, the bound from axiom A3, DiscrepancyMeasure(h) ≤ $\max_k |\hat{F}_h(k)|$, scales homogeneously of degree 1 under positive tropical rescaling, so the inequality is preserved exactly under $\lambda > 0$ (no slack opens or closes).

Rationale

This conjecture directly tests axiom A3 (DiscrepancyMeasure ≤ max Fourier coefficient) jointly with axiom A2 (invertibility of TropicalFourierTransform on the chosen class). Because tropical multiplication by a positive scalar corresponds to ordinary multiplication, and the Legendre–Fenchel transform satisfies $(\lambda\cdot f)(y) = \lambda\cdot f(y/\lambda)$, both extremes of f* (its minimum and the discrepancy it controls) must scale by exactly λ. Verifying this 1-homogeneity is a clean sanity check that the framework's notion of MinimalFourierCoefficient behaves as a genuine 'norm-like' invariant compatible with the tropical semiring's scalar action, and that the A3 bound is not an artifact of a particular scale (relevant for transferring lower bounds in tropical/min-plus circuit complexity, where rescaling weights is a standard reduction).

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [1207.2443v2] Tropical Teichmuller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [1406.3065v2] Lower Bounds for Tropical Circuits and Dynamic Programs - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21db269d.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21db269d.py", line 71, in run_trial
    f = [max([random.randint(-10, 10) * x[i] + random.randint(-10, 10) for i in range(d)]) for _ in rang
                                        ^
NameError: name 'x' is not defined
```

Judge reasoning

The test crashed with a NameError ('x' is not defined') before any trials executed, so no data was produced to evaluate the pre-registered support or falsification conditions. `|` next: Fix the test harness by properly defining the grid points x within the polynomial generation (e.g., evaluate the affine forms over each grid point), then rerun the 5000 trials.

Permanent-Variance of NW Design Matrix Predicts PRG Bias

- **Verdict**: INCONCLUSIVE
- **Bridge**: Combinatorics of 0/1 matrix permanents (Ryser-formula variance over random row-sign perturbations) × Nisan-Wigderson pseudorandom generator output bias against small DNF distinguishers
- **Recorded**: 2026-04-27 02:42 UTC
- **Entry ID**: 92f05cc72075

Statement

Let D be an (ell,a)-NW design on n variables encoded as an m x n 0/1 incidence matrix M (rows = subsets S_i, columns = universe). Define V(M) := log2(1 + Var_{eps in {+/-1}^n}[perm(M_eps)]) / m, where M_eps is M with column j scaled by eps_j and perm is the 0/1 permanent on the (m x m) submatrix induced by the m heaviest columns. We conjecture that for every hard truth table f : {0,1}^ell -> {0,1} of formula-leaf-count L(f) >= 2^{ell/3}, the maximum bias of NW_{D,f} against width-w DNF distinguishers satisfies bias(NW_{D,f}, w) <= 2^{-c * V(M) * m / w} for an absolute constant c > 0, so V(M) is a constructive lower bound certificate for PRG quality.

Rationale

Permanent variance over sign perturbations measures how 'spread' the design's row-overlap pattern is, which is exactly the combinatorial quantity controlling pairwise-intersection cancellation in the NW hybrid argument; existing analyses use only |S_i cap S_j| <= a, but the variance aggregates higher-order interference. Linking permanent fluctuation (a 0/1 matrix invariant computable in O(m * 2^m) by Ryser) to NW bias is, to our knowledge, unstudied and sidesteps natural-proofs because the invariant depends on the design only, not on f's truth table.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1311.6531v3] Brains and pseudorandom generators - [2412.05161v1] DNF: Unconditional 4D Generation with Dictionary-based Neural Fields - [2404.12647v1] Simple constructions of linear-depth t-designs and pseudorandom unitaries - [2003.09703v1] Variance function of boolean additive convolution - [2012.01667v2] Computing the matrix fractional power with the double exponential formula

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08ccc640.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08ccc640.py", line 66, in run_trial
    perm_var = permanent([[M_heaviest[i][j] * M_heaviest[j][i] for j in range(m)] for i in range(m)])
                                              ~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so no support fraction or violation ratio could be computed against the pre-registered criterion. | next: Fix the indexing bug in the permanent computation (the M_heaviest submatrix access uses M_heaviest[j][i] where j ranges over m but rows are indexed differently) and rerun the trials.

Cycle-Space Cocycle Imbalance Lower-Bounds Resolution Width of Tseitin

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic graph theory: GF(2) cycle space and edge-boundary cocycle imbalance of expander graphs × Resolution refutation width of Tseitin formulas on bounded-degree expanders
- **Recorded**: 2026-04-27 03:18 UTC
- **Entry ID**: `09c50eb1a365`

Statement

Let G=(V,E) be a connected 3-regular graph and let T(G,c) be the Tseitin formula for an odd charge c:V->{0,1}. Define the cocycle imbalance I(G) = min over nonempty S subset V of |E(S, V§)| / min(|S|, |V§|), and let w(T(G,c)) denote the minimum width of any resolution refutation of T(G,c). The conjecture is the linear lower bound w(T(G,c)) >= floor(I(G) * min(|S|*, |V§*|) / 2) + 1, where S* is the cocycle-imbalance minimizer; equivalently, resolution width is at least half the minimum odd-charge edge-boundary plus one.

Rationale

Ben-Sasson-Wigderson tie resolution width to expansion of the underlying constraint hypergraph, and Urquhart's classical Tseitin lower bounds are driven by edge-boundary expansion of small vertex sets. Reformulating expansion as a GF(2) cocycle imbalance (the dual of cycle-space sparsity) gives a coordinate-free invariant that should sharpen the constant in width lower bounds, and falsification on a single 3-regular graph would refute a clean algebraic strengthening of the BSW bound.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1411.3958v3] Edge-state enhanced transport in a 2-dimensional quantum walk - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0804.4433v1] SPACE: the SPectroscopic All-sky Cosmic Explorer - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation

Empirical Test

- exit code: `124`, elapsed: `180.08s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any RESULT line, so the pre-registered support condition over ≥210 instances was never evaluated. | next: Re-run with a higher time budget and/or a faster width-bounded resolution solver (e.g., restrict to n ≤ 14 first, parallelize per-instance, and cache I(G) computations) to actually collect the 210 data points.

Halton Star-Discrepancy of Clause-Sign Embedding Bounds DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Quasi-Monte Carlo low-discrepancy sequence theory (Halton/van der Corput star-discrepancy) × DPLL search-tree leaf count on random 3-CNF instances
- **Recorded**: 2026-04-27 03:49 UTC
- **Entry ID**: `c4e28e9508f6`

Statement

Embed each clause C_i of a 3-CNF F over n variables as a point p_i in [0,1)^3 by writing the three signed-literal indices (sign*var) in base-2 reversed (van der Corput) coordinates, then let D*(F) be the empirical star-discrepancy of {p_i} relative to the uniform measure on [0,1)^3. Conjecture: for unsatisfiable random 3-CNF at clause density alpha=4.267, the DPLL leaf count L(F) satisfies log2 L(F) >= c * n * D*(F) - O(log n) with c = 1/2, and equivalently log2 L(F) <= C* n * D*(F) + O(log n) with C = 4. A single instance with log2 L(F) < 0.4* n * D*(F) - 5 (n<=20) falsifies the lower bound.

Rationale

Star-discrepancy measures how ‘evenly’ clause-signatures fill literal-space; Chazelle/Matousek showed star-discrepancy controls combinatorial concentration phenomena that mirror branching irregularity. Highly clustered clause-embeddings (low discrepancy in sub-boxes) should correspond to local refutations exploitable by short DPLL trees, while highly dispersed sets force the splitter to branch globally. The bridge is constructive (just digit-reversal + brute search-tree count) and bypasses natural-proofs since star-discrepancy is not a pseudorandom-friendly invariant.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1209.3611v2] A van der Corput-type algorithm for LS-sequences of points - [2411.14979v2] Fundamentals and applications of Van der Waals magnets in magnon spintronics - [1908.01624v1] Learned Clause Minimization in Parallel SAT Solvers - [1402.4455v1] Symbiosis of Search and Heuristics for Random 3-SAT - [0506069v1] A generating function method for the average-case analysis of DPLL

Empirical Test

- exit code: `124`, elapsed: `180.03s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any data, so neither the support condition nor the falsification condition could be evaluated. | next: Reduce the workload (e.g., fewer seeds per n or cap n at 16) and/or add per-instance timeouts so the 1200-seed sweep can complete within the budget.

Erdős–Ko–Rado Shadow Defect Lower-Bounds Disjointness Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal set theory via Kruskal–Katona shadow profiles and EKR-type intersecting family deficits × Discrepancy-based randomized two-party communication complexity of partial set-disjointness matrices
- **Recorded**: 2026-04-27 04:23 UTC

- **Entry ID**: `78c33f7389a9`

Statement

For an n×n ±1 matrix M_F constructed from a k-uniform set family F⊆{0,1}^n (rows = members of F, columns = members of F, entry = +1 if disjoint, -1 if intersecting), let s(F) be the Kruskal–Katona shadow defect, defined as |∂F| − KK_min(|F|,k−1) where ∂F is the (k−1)-shadow and KK_min is the colex minimum, and let δ(F) be the spectral discrepancy ‖M_F‖_∞→1 / (|F|^2). We conjecture that for every k-uniform F with 2≤k≤n/2, δ(F) ≥ c · s(F) / (|F| · binomial(n,k−1)) for a universal constant c≥1/8, with equality approached only by colex-initial (EKR-extremal) families. A single (n,k,F) with δ(F) < (1/8) · s(F) / (|F| · C(n,k−1)) refutes it.

Rationale

Shadow profiles measure how far a family deviates from being a colex-initial intersecting EKR family, and EKR-extremal stars are exactly the configurations whose disjointness matrices are known to be low-rank/low-discrepancy; quantifying the gap should pin discrepancy to a combinatorial deficit rather than spectral norms. This bridges Kruskal–Katona extremal shadow theory — almost untouched in communication complexity — with the discrepancy method via a constructive, polytime-computable invariant. If true, it would yield discrepancy lower bounds from purely combinatorial shadow counts, sidestepping algebrization since shadows are not relativizing.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1809.03967v2] More Cases Where the Kruskal-Katona Bound is Tight - [2005.06999v1] Tight bounds for Katona's shadow intersection theorem - [2305.10152v1] On the extremal families for the Kruskal–Katona theorem - [1408.1376v2] Factorization Norms and Hereditary Discrepancy - [1208.6015v1] The spectral function of a first order elliptic system

Empirical Test

- exit code: `124`, elapsed: `180.10s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out at 180s (returncode 124) before producing any data, so neither the support nor falsification condition can be evaluated. The critic's CHALLENGE stands. | next: Reduce the search space (e.g., n∈{8..12}, k∈{2,3}, 50 random samples per cell) and optimize the spectral discrepancy computation (use ‖M‖_∞→1 via sign-vector enumeration only for small |F|, or a Grothendieck-style SDP relaxation) to fit within the time budget.

Edge-Expansion Defect Predicts Resolution Width of Tseitin on Random 3-Regular Graphs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete differential geometry via vertex-weighted Cheeger defect: the gap between the bottleneck edge-isoperimetric ratio h(G) and its second-eigenvalue Cheeger upper bound 2*sqrt(d*lambda_2(L)) on the unnormalized graph Laplacian × Resolution refutation width of Tseitin formulas T(G,c) on small 3-regular graphs with odd total charge
- **Recorded**: 2026-04-27 04:57 UTC
- **Entry ID**: `c3c56f86d197`

Statement

Let G be a connected 3-regular graph on n vertices with odd-charge assignment c, let h(G) = min_{|S|<=n/2} |E(S, V§)|/|S| be its edge-isoperimetric number, and let beta(G) = 2*sqrt(3*lambda_2(L_G)) be the Alon-Milman/Cheeger upper bound. Define the Cheeger defect Delta(G) = beta(G) - h(G) >= 0. We conjecture that the minimum resolution refutation width w(T(G,c)) satisfies (1/3)*h(G)*(n - Delta(G)*n/beta(G)) - 2 <= w(T(G,c)) <= h(G)*n/2 + 2 for every odd-charge c, and in particular the *defect-corrected* lower bound h(G)*(1 - Delta(G)/beta(G))*n/3 - 2 is tight up to constants on random 3-regular graphs with high probability.

Rationale

Ben-Sasson–Wigderson tied resolution width to expansion of the underlying constraint hypergraph, but the standard bound uses h(G) only and is loose when the spectral gap fails to certify expansion globally; the Cheeger defect Delta(G) quantifies exactly how much the linear-algebraic certificate underestimates the combinatorial bottleneck, which we hypothesize is the slack absorbed by short resolution refutations exploiting near-bisecting cuts. Because both h(G) and lambda_2(L_G) are computable for n <= 20 by brute-force enumeration over subsets and a 20x20 eigensolver, the bridge is fully constructive. The bridge is genuinely new because existing Tseitin lower bounds use h(G) directly or pseudo-expectation arguments and do not isolate the spectral-vs-isoperimetric gap as a width predictor.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0908.1772v5] Some Probabilistic Results on Width Measures of Graphs - [0407622v3] A Study of the Composition of Ultra High Energy Cosmic Rays Using the High Resolution Fly's Eye - [1712.06904v3] Rigidity for the isoperimetric inequality of negative effective dimension on weighted Riemannian manifolds - [2111.02944v1] A sharp isoperimetric inequality in metric measure spaces with non-negative Ricci curvature

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 131, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 102, in run_trial
    lambda_2_G = compute_lambda_2(G)
                 ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 54, in compute_lambd
    eigenvector = power_iteration(A)
                  ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 44, in power_iterati
    x /= math.sqrt(sum(x[i] ** 2 for i in range(n)))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

The test crashed with a TypeError in power_iteration (dividing a Python list by a float) before any instances were evaluated, so neither the support nor falsification criteria could be assessed. | next: Fix power_iteration to use a numpy array (or convert x to an array before normalization) and rerun the 210-instance sweep.

Average-Case DPLL Runtime Linked to Random Matrix Spectra

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × DPLL Algorithm Runtime
- **Recorded**: 2026-04-27 06:23 UTC
- **Entry ID**: `7ac5fad5a356`

Statement

For random 3-CNF formulas with n variables, the expected DPLL runtime is exponentially bounded by the largest eigenvalue λ_max of the clause adjacency matrix, i.e., $E[T(n)] = \Theta(2^{\lambda_{max} * n})$.

Rationale

Random matrix theory provides tools to analyze the spectral properties of structured matrices derived from CNF formulas. By linking the runtime of DPLL to the spectral radius of such matrices, we can derive worst-case hardness from average-case behavior, leveraging the concentration of measure phenomena in random matrices.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5978c4e7.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5978c4e7.py", line 76, in run_trial
    lambda_max = largest_eigenvalue(adjacency_matrix)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5978c4e7.py", line 51, in largest_eigen
    v /= math.sqrt(sum(x**2 for x in v))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

Test crashed during eigenvalue computation, preventing data collection to assess support/falsification | next: Fix the eigenvector normalization code to handle list operations properly and rerun experiments

---

Betti Numbers and Resolution Width in 3-CNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology × Resolution Width
- **Recorded**: 2026-04-27 07:04 UTC
- **Entry ID**: `6bc88662420d`

Statement

For a random 3-CNF instance with n variables, the first Betti number of the clause-induced simplicial complex is Θ(log n) if and only if the resolution width is Θ(log n).

Rationale

Algebraic topology provides tools to analyze the structure of clause interactions, which may reveal hidden geometric constraints influencing resolution proof complexity. Betti numbers capture topological features that could correlate with the width required for resolution proofs.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1009.4130v3] Limit theorems for Betti numbers of random simplicial complexes - [2003.08066v2] Law of large numbers for Betti numbers of homogeneous and spatially independent random simplicial complexes - [1501.03759v1] Erratum: Limit theorems for Betti numbers of random simplicial complexes - [2110.02458v3] Magnitude homology of graphs and discrete Morse theory on Asao-Izumihara complexes - [2306.01534v3] The magnitude homology of a hypergraph

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5ea9db14.py", line 142, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5ea9db14.py", line 95, in run_trial
    variables = random.sample(variable_set, 3)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError (random.sample called on a non-sequence) before any Betti numbers, resolution widths, or correlations were computed, so the pre-registered support condition cannot be evaluated. | next: Fix the bug by passing a sorted list (e.g., random.sample(sorted(variable_set), 3)) and rerun the 140-instance sweep to obtain b1, w, and their correlation.

Sandpile-Group Order of Lifted IP Bipartite Graph Bounds Decision Tree Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic graph theory via the critical (sandpile) group K(G) and Kirchhoff's Matrix-Tree theorem applied to bipartite communication graphs (rare in CS lower bounds; cf. Bacher–de la Harpe–Nagnibeda, Biggs) × Deterministic communication complexity of Raz–McKenzie / Göös–Pitassi-style lifted Boolean functions f ∘ IP_b^n with the inner-product gadget
- **Recorded**: 2026-04-27 07:15 UTC
- **Entry ID**: `69ad4d50946e`

Statement

Let $f:\{0,1\}^n\to\{0,1\}$ have exact decision-tree depth $d=D(f)$, let $b\in\{2,3\}$ be the IP-gadget block size, and let $M(f,b)\in\{0,1\}^{2^{bn}\times 2^{bn}}$ be the lifted communication matrix with $M[x,y]=f(\langle x_1,y_1\rangle,\ldots,\langle x_n,y_n\rangle)$ over GF(2). Form the bipartite graph $G(f,b)$ with rows on one side, columns on the other, and an edge exactly where $M[x,y]=1$; let $\tau(G)$ denote its spanning-tree count (Kirchhoff determinant of any reduced Laplacian, with $\tau=0$ if disconnected and τ a product over components otherwise). Conjecture: for every non-constant f with $n\le 3$ and $b\in\{2,3\}$, $\log_2 \tau(G(f,b)) \ge d \cdot 2^{bn-2}$, and equality is achieved (within $+b$) precisely when f is the parity (XOR_n) function.

Rationale

Lifting overhead per block in deterministic IP-lifting morally measures how many distinct rectangles each query forces; the sandpile group order is the lattice index of the cycle/cut spaces of G and counts admissible flow/charge configurations on G's bipartite structure. Linking $\tau(G)$ to $D(f)\cdot$(gadget block area) plants a query-to-communication bridge through pure graph algebra, sidestepping algebrization since spanning-tree counting is not a relativizing arithmetic primitive. Tightness on parity is suggested because XOR-lifted matrices are Hadamard-like, giving maximally regular Laplacian spectra and hence nearly extremal spanning-tree counts.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1906.08515v1] An overview on the bipartite divisor graph for the set of irreducible character degrees - [1705.07519v1] Sandpile Groups of Random Bipartite Graphs - [1904.12209v2] Sandpile monomorphisms and limits - [1012.2478v2] Packing spanning trees and the k-tree protocol - [0904.3087v4] Distributed Maintenance of Anytime Available Spanning Trees in Dynamic Networks

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb3bd595.py", line 134, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb3bd595.py", line 96, in run_trial
    query = memoized_recursive_query_enumeration(lambda x: f[x], n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb3bd595.py", line 21, in memoized_recu
    @lru_cache(None)
     ^^^^^^^^^
NameError: name 'lru_cache' is not defined
```

Judge reasoning

The test crashed with a NameError ('lru_cache' is not defined) before producing any data, so no support fraction or slack values were computed. Per the rules, a crashed test yields INCONCLUSIVE. | next: Fix the test by importing lru_cache from functools (add 'from functools import lru_cache') and rerun the full enumeration over $n\in\{1,2,3\}$, $b\in\{2,3\}$.

Doob Martingale Variance Gap Predicts Worst-Case DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Doob martingale concentration / Efron-Stein variance decomposition on random CSP exposures × DPLL search-tree depth on random 3-CNF (average-case to worst-case lift)
- **Recorded**: 2026-04-27 07:44 UTC
- **Entry ID**: `6487f7d68c31`

Statement

Let F ~ F(n, m=4n) be a random 3-CNF and let X(F) = log2(DPLL_leaves(F)) under fixed unit-propagation + first-unassigned variable order. Expose clauses one at a time to form the Doob martingale M_i = E[X | C_1,…,C_i] and let V(F) = sum_i (M_i - M_{i-1})^2 be its realized quadratic variation. Then for every n in [10,20] there exist constants c1,c2 > 0 (independent of n) with c1 * sqrt(V(F)) * log n <= max_{F' agreeing with F on all but one clause} X(F') - E[X] <= c2 * sqrt(V(F)) * log n with probability >= 1 - 1/n; equivalently, the average-case Doob variance V(F) tightly predicts the one-clause-perturbation worst-case depth gap.

Rationale

Efron-Stein variance bounds give average-case concentration, but the lower-tail (one-clause-resampling worst-case) is rarely tied to the same martingale variance — yet Impagliazzo-Wigderson-style hardness amplification suggests average-case fluctuation should control nearby worst-case spikes. If the two-sided bound holds, an average-case lower bound on V translates into a worst-case DPLL depth bound via a single resampling step, a concrete avg-to-worst lift on a small computable scale.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1411.0186v2] Algorithmic randomness for Doob's martingale convergence theorem in continuous time - [1408.3470v1] Efron-Stein Inequalities for Random Matrices - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [0506069v1] A generating function method for the average-case analysis of DPLL - [2002.02797v4] Variational Depth Search in ResNets

Empirical Test

- exit code: `0`, elapsed: `0.04s`


```
{"TRIAL": {"seed": 11, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 23, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 37, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 53, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 71, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```


Judge reasoning

The metric_value is exactly 0 with zero variance across all 5 seeds and only 6 instances per seed, indicating the test likely checked a trivially-satisfiable existential bound rather than the pre-registered OLS R^2/slope/window-fraction criterion. No evidence of pooled R^2, fitted slope, or per-n window fractions was reported, so the support condition is not unambiguously met. | next: Re-run with seeds 1..200 across n in {10,12,14,16,18,20} and explicitly compute pooled OLS regression of log(gap

---

Newton-Inequality Defect of SAT Generating Polynomial Bounds DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Pólya–Schur theory of real-rooted polynomials and Newton's inequalities (multiplier sequences, log-concavity defects of integer coefficient polynomials) × Average-case DPLL search-tree depth on random 3-CNF and its direct-product worst-case lift
- **Recorded**: 2026-04-27 08:18 UTC
- **Entry ID**: `4907d4d1fa3e`

Statement

For a 3-CNF F on n variables, define the solution generating polynomial Phi_F(z) = sum_{k=0}^n a_k(F) z^k where a_k(F) = #{x in {0,1}^n : F(x)=1 and weight(x)=k}, and let the Newton-inequality defect be delta(F) = max_{1<=k<=n-1} max(0, a_{k-1} a_{k+1} / a_k^2 - (k-1)(n-k+1)/(k(n-k+1))) with the convention 0/0 = 0. Conjecture: there exist constants c1, c2 > 0 (independent of n) such that for random satisfiable 3-CNF F at clause density alpha in [3.6, 4.0], the expected average-case DPLL leaf count satisfies E[log2 leaves_DPLL(F)] >= c1 * n * log2(1 + delta(F)) - c2 * sqrt(n), and the k-fold disjoint-variable direct product F^{(k)} satisfies leaves_DPLL(F^{(k)}) >= (1 + delta(F))^{c1 * k * n} for every static branching order. The conjecture is falsified by exhibiting a single F (with n<=20) whose measured leaf count is below c1 * n * log2(1 + delta(F)) - c2 * sqrt(n) for c1 = 0.25, c2 = 4.

Rationale

Pólya–Schur theory says that polynomials with only real (negative) roots satisfy Newton's inequalities exactly; the defect delta quantifies how far Phi_F is from being a multiplier-stable polynomial. Real-rooted Phi_F corresponds to clusters of solutions arranged like an independent-coordinate product distribution, which DPLL prunes efficiently; conversely, large delta signals pinned correlations between Hamming-weight strata of the solution set, which force DPLL backtracks. Direct product then transports this average-case structural obstruction to a worst-case depth lower bound in the spirit of Impagliazzo–Wigderson hardness amplification.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1907.08306v1] A Polynomial Time Algorithm for Log-Concave Maximum Likelihood via Locally Exponential Families - [2004.07220v2] Log-Concave Polynomials IV: Approximate Exchange, Tight Mixing Times, and Near-Optimal Sampling of Forests - [1604.00993v1] The 'Core' of Symmetric Homogeneous Polynomial Inequalities of Degree Four of Three Real Variables - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [0803.0018v5] Another approach to decide on real root existence for univariate Polynomials, and a multivariate extension for 3-SAT

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60f182d0.py", line 160, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60f182d0.py", line 103, in run_trial
    clauses = generate_random_satisfiable_3cnf(n, alpha)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60f182d0.py", line 88, in generate_rand
```

```
    if any(evaluate_formula(clauses, dict(enumerate([random.randint(0, 1) for _ in range(n)]))) for _ i
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60f182d0.py", line 88, in <genexpr>
    if any(evaluate_formula(clauses, dict(enumerate([random.randint(0, 1) for _ in range(n)]))) for _ i
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60f182d0.py", line 29, in evaluate_form
    return all(all((assignment[var] + lit) % 2 == 0 for var, lit in clause.items()) for clause in formula
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_60f182d0.py", line 29, in <genexpr>
    return all(all((assignment[var] + lit) % 2 == 0 for var, lit in clause.items()) for clause in formula
                                                                    ^^^^^^^^^^^^
AttributeError: 'list' object has no attribute 'items'
```

Judge reasoning

The test crashed with an AttributeError in evaluate_formula (clauses stored as lists rather than dicts), so no data was produced and neither the support nor falsification criteria can be evaluated. | next: Fix the clause data-structure bug in evaluate_formula/generate_random_satisfiable_3cnf (use a consistent representation, e.g., list of (var, lit) tuples) and rerun the full 360-instance pre-registered protocol.

Möbius Function of Minterm Lattice Bounds Indexing-Lifted Communication

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Topological combinatorics (Möbius function and order-complex Euler characteristic on the meet-closure lattice of minterm supports, in the spirit of Rota and Stanley) × Deterministic two-party communication complexity of Indexing-gadget-lifted monotone functions (Göös–Pitassi–Watson lifting), with deterministic decision-tree depth as the lifted-side proxy
- **Recorded**: 2026-04-27 08:47 UTC
- **Entry ID**: `b63c45216b7a`

Statement

Let $f:\{0,1\}^n\to\{0,1\}$ be monotone with prime-implicant supports $S_1,\dots,S_k\subseteq[n]$; let L_f be the meet-closure lattice generated by $\{S_i\}$ together with an adjoined top $\hat{1}=[n]$ and bottom $\hat{0}=\emptyset$, and let $\mu_f:=|\mu_{L_f}(\hat{0},\hat{1})|$ be its Möbius number. Then for all $n\leq 20$ the deterministic decision-tree depth satisfies $D^{dt}(f) \geq \lceil \log_2(1+\mu_f)\rceil$, and consequently the Göös–Pitassi–Watson Indexing-gadget lifting yields $D^{cc}(f\circ IND_b) = \Omega((\log b)\cdot\log_2(1+\mu_f))$. A single monotone f with $\lceil\log_2(1+\mu_f)\rceil > D^{dt}(f)$ refutes the conjecture.

Rationale

Lifting theorems convert decision-tree depth to communication, so any combinatorial invariant that lower-bounds $D^{dt}(f)$ automatically lower-bounds lifted communication. The Möbius number of the minterm closure lattice measures how much non-Boolean-lattice structure the accepting set carries; intuitively, each query in a decision tree can only halve the alternating Möbius sum on a chain, so depth must dominate log of $|\mu|$. This bridges topological combinatorics to communication lower bounds, a pairing with no prior occurrences I am aware of.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0710.4339v1] Heavy-Quark Masses from the Fermilab Method in Three-Flavor Lattice QCD - [1111.0981v1] Form factors for B to Kll semileptonic decay from three-flavor lattice QCD - [1810.08668v2] Parity Decision Tree Complexity is Greater Than Granularity - [1605.01142v2] Separations in communication complexity using cheat sheets and information complexity - [2001.02144v1] Lifting with Simple Gadgets and Applications to Circuit and Proof Complexity

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6637e14.py", line 125, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6637e14.py", line 71, in run_trial
    dnf = generate_random_dnf(n, k, implicant_size)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6637e14.py", line 31, in generate_rand
    if all(len(set(a).intersection(b)) == 0 for b in implicants):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6637e14.py", line 31, in <genexpr>
    if all(len(set(a).intersection(b)) == 0 for b in implicants):
                   ^
NameError: name 'a' is not defined
```

Judge reasoning

The test crashed with a NameError in generate_random_dnf (undefined variable 'a' in a generator expression) before any instances were evaluated, so no data was produced for either the support or falsification criterion. | next: Fix the generator-expression bug in generate_random_dnf (likely should iterate over candidate 'a' against existing implicants 'b') and rerun the 400-instance sweep.

Noise-Stability Plateau at rho=1/3 Bounds DPLL Leaves on 3-CNF

- **Verdict**: INCONCLUSIVE
- **Bridge**: Fourier analysis of Boolean functions (noise stability Stab_rho and Bonami-Beckner noise operator T_rho) × DPLL search-tree leaf count on random 3-CNF formulas
- **Recorded**: 2026-04-27 09:18 UTC
- **Entry ID**: 3385dd792f63

Statement

For a 3-CNF formula F on n variables with indicator function f_F:{-1,1}^n -> {0,1} (1 iff assignment satisfies F), let S(F) := Stab_{1/3}(f_F) - (E[f_F])^2 = sum_{|T|>=1} (1/3)^{|T|} hatf(T)^2. We conjecture that for every unsatisfiable 3-CNF F with m clauses on n<=20 variables, the number L(F) of leaves explored by the standard unit-propagating DPLL with fixed lexicographic branching satisfies L(F) >= 2^{n} * S(F) / (3 * m), and equivalently for satisfiable F we get L(F) <= 2^{n} * (1 - 9*S(F)/m). One satisfiability instance violating either inequality refutes the conjecture.

Rationale

At noise rate 1/3 each clause's local indicator (a degree-3 Boolean function on three variables) decays by exactly (1/3)^3=1/27 per level, matching the branching factor of DPLL on a 3-CNF; thus the rho=1/3 noise stability isolates the ‘clause-coupled’ Fourier mass that DPLL must shatter. Kahn-Kalai-Linial-style influence bounds show that each clause contributes at most O(1/n) to total influence, so summing Stab_{1/3} over clauses should track tree size up to the clause-density factor m. This bridges hypercontractive Fourier weight with concrete branching cost, a connection rarely tested empirically.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2003.09703v1] Variance function of boolean additive convolution - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [1112.2313v2] QBF-Based Boolean Function Bi-Decomposition - [1402.4455v1] Symbiosis of Search and Heuristics for Random 3-SAT - [0710.0805v3] On the Satisfiability Threshold and Clustering of Solutions of Random 3-SAT Formulas

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_689eea86.py", line 98, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_689eea86.py", line 62, in run_trial
    f_F = {tuple([random.choice([-1, 1]) for _ in range(n)]): evaluate_formula(f_F, assignment) for ass
1, 1], repeat=n)}
                                                                                          ^^^
UnboundLocalError: cannot access local variable 'f_F' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so no instances were evaluated and neither the support nor falsification condition can be assessed. | next: Fix the test harness bug (initialize f_F properly before constructing the assignment-to-evaluation dictionary) and rerun the full 1050-instance sweep.

Brooks Quasimorphism Magnitude Lower-Bounds Formula Depth via Barrington Lifts

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric group theory: Brooks counting quasimorphisms on the free group F_2 (bounded cohomology H_b^2(F_2;R), Bestvina–Fujiwara) × NC^1 / Boolean formula depth via Barrington’s width-5 S_5 branching program lifted to F_2
- **Recorded**: 2026-04-27 09:53 UTC
- **Entry ID**: `5013aa48b582`

Statement

For a Boolean formula φ of syntactic depth d over n variables, build Barrington’s canonical width-5 ground BP B_φ (length 4^d) and lift each S_5 factor to F_2=⟨a,b⟩ via the surjection a↦(1 2 3 4 5), b↦(1 2), giving a word w_φ(x)∈F_2 of length 4^d for every input x∈{0,1}^n. Let τ(w):=#{occurrences of the reduced subword `aba` in w}−#{occurrences of `a^{-1}b^{-1}a^{-1}`} be the Brooks counting quasimorphism on F_2 with seed `aba`. Conjecture: there exist absolute constants c_1,c_2>0 such that for every φ with d≥2, c_1·2^d ≤ max_{x∈{0,1}^n} |τ(w_φ(x))| ≤ c_2·2^d, so depth(φ)=log_2 max_x|τ(w_φ(x))|±O(1).

Rationale

Barrington's recursion nests commutators d-deep, and in the free group F_2 nested commutators are precisely the words on which Brooks quasimorphisms grow linearly per nesting level (Bavard duality, scl-detection). Solvable groups have bounded commutator length so this signal collapses, but F_2 — the universal Barrington target — preserves it, suggesting a single F_2-word invariant can certify NC^1 depth orthogonally to communication, polynomial-method, or approximation-based lower bounds. No prior work I know of pairs Brooks quasimorphisms with Barrington BPs.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1911.10855v2] $\hat{G}$-invariant quasimorphisms and symplectic geometry of surfaces - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group - [1710.02392v2] The cup product of Brooks quasimorphisms - [1901.01301v1] Second bounded cohomology and WWPD - [0701214v2] Graphs, free groups and the Hanna Neumann conjecture

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_252ab5e3.py", line 130, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_252ab5e3.py", line 103, in run_trial
    triples = barrington_construction(formula)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_252ab5e3.py", line 28, in barrington_co
    return stack[0]
           ~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in `barrington_construction` before any data was collected, so neither the support nor the falsification condition can be evaluated. | next: Fix the Barrington construction (likely a stack underflow when handling leaf/literal gates or odd recursion base cases) and rerun the pre-registered protocol across d∈{2,3,4,5}.

Hadamard-Code Distance Defect Predicts NW-PRG Distinguisher Advantage

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coding theory of binary Reed-Muller / Hadamard codes (minimum-distance defect of the design's combinatorial incidence code) × Nisan-Wigderson pseudorandom generator output bias against small parity-AND distinguishers
- **Recorded**: 2026-04-27 10:22 UTC
- **Entry ID**: `1a5be3baec68`

Statement

Let D be an (m,l,a)-NW design on n=|S_i| sets and let M_D in {0,1}^{n x m} be its incidence matrix; let d(D) be the minimum Hamming distance between any two distinct rows of M_D and let d* = l - a be the design’s pairwise-overlap upper bound on agreement. For every hard predicate f:{0,1}^l -> {0,1} of correlation eps with degree-1 GF(2) polynomials, the maximum distinguishing advantage of any AND-of-c-parities distinguisher against NW_{D,f} satisfies adv(NW_{D,f}) <= c * eps * 2^{-(d(D) - 2(l-d*))/2}, and this bound is tight up to a constant for at least 30% of random hard f at every (m,l,a,n<=20).

Rationale

NW analysis classically pays a 2^{c*a} factor for c-parity distinguishers via the design’s overlap a; recasting M_D as a binary code lets the* actual* row-distance d(D) — typically much larger than the worst-case 2(l-a) bound — control distinguisher leakage through a Plotkin-style inner-product argument. Coding-theoretic distance has been almost entirely absent from NW PRG analyses (which use overlap, not minimum distance), and the gap d(D) - 2(l-a) is exactly the slack that explains why concrete designs beat their generic guarantee. The conjecture is falsifier-friendly: any single (D,f,c) tuple where the empirical advantage exceeds the RHS by more than a small constant kills it.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2106.11085v3] On the Maximal Monotone Operators in Hadamard Spaces - [1003.4003v1] A Fourier-analytic Approach to Counting Partial Hadamard Matrices - [1003.4001v1] On the Asymptotic Existence of Hadamard Matrices - [1311.6531v3] Brains and pseudorandom generators - [2410.08073v3] Efficient Quantum Pseudorandomness from Hamiltonian Phase States

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c9b9fc0.py", line 120, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c9b9fc0.py", line 95, in run_trial
    advantage = nw_distinction(design, lambda x: sum(x) % 2 == parity[0], parity)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c9b9fc0.py", line 72, in nw_distinctio
    if predicate(seed) == parity[i]:
                          ~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification conditions can be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the nw_distinction harness so the parity vector is sized to match the number of distinguisher queries (length c), then re-run the enumeration over (m,l,a,n<=20) with >=5 seeds.

Asymptotic-dimension lower bound for the indexing function via controlled covers matches its log-depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); coarse geometry / Roe algebras / metric (Roe) cohomology × Formula depth (over the De Morgan basis, equivalently NC^1 circuit depth) of the addressing/indexing function IND_k : {0,1}^{k + 2^k} -> {0,1}, which selects data bit x_a where a is encoded by the k address bits; IND_k has known formula depth Θ(k) and is the canonical KW-easy test instance for any proposed depth measure.
- **Recorded**: 2026-04-27 10:37 UTC
- **Entry ID**: `e871be18b26c`

Statement

For every k ≥ 1, the KW-metric space (X_{IND_k}, d_{IND_k}) admits no controlled cover of diameter R = 2^{k-1} − 1 with multiplicity m ≤ k, and consequently asdim(X_{IND_k}) ≥ k − O(1); together with axiom A1 this yields κ(IND_k) ≥ k − O(1) = Ω(log N) where N = k + 2^k is the input length, matching the true depth Θ(k) up to a constant. Equivalently: there exists an explicit coarse 1-cocycle c_k ∈ HX^1(X_{IND_k}) and a Roe-controlled operator T_k of propagation 2^{k} with |⟨c_k, T_k⟩| ≥ 2^{Ω(k)}, certifying a super-constant κ on this family.

Rationale

This sub-conjecture stress-tests axiom A1 (Coarse-KW link) on a function whose true depth is precisely known: if A1 holds, the controlled-cover multiplicity must witness the Karchmer-Wigderson protocol's depth, so on IND_k we MUST recover k − O(1). It also exercises axiom A5 (Roe-index rigidity) by demanding an explicit cocycle/operator pair realizing the trace pairing — IND_k is the simplest function whose KW-tree is a complete binary tree, so the address-bit projections give natural candidates for c_k. Failure on IND_k would falsify A1; success calibrates the constants in A1 before attempting the Andreev / element-distinctness Ω(log^2 n) targets.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b843142.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b843142.py", line 143, in run_trial
    cover = controlled_cover(R, k)
            ^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b843142.py", line 41, in controlled_co
    for i in range(1 << (log2_floor(node) + 1)):
                         ^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b843142.py", line 25, in log2_floor
    return math.floor(math.log2(n))
```

```
                         ^^^^^^^^^^^^
ValueError: math domain error
```

Judge reasoning

The test crashed with a ValueError (math domain error in log2_floor due to a non-positive argument) before producing any slope, trace-pairing, or baseline data, so none of the pre-registered support conditions can be evaluated. | next: Guard log2_floor against $n \leq 0$ (e.g., handle the root/zero node explicitly) and rerun the controlled-cover and trace-pairing measurements for $k \in \{2,3,4\}$ with the 20-seed random baseline.

Permanent of Sensitive-Boundary Bipartite Matrix Bounds Lifted Indexing CC

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Permanents of bipartite 0/1 biadjacency matrices via Ryser inclusion-exclusion enumeration of perfect matchings (algebraic combinatorics on the Boolean cube; rarely deployed in decision-tree / communication lower bounds) × Deterministic communication complexity D^cc(f ∘ IND_b) of Indexing-lifted Boolean functions, accessed through decision-tree depth Q^dt(f) via the Raz-McKenzie / Göös-Pitassi-Watson lifting theorem
- **Recorded**: 2026-04-27 10:47 UTC
- **Entry ID**: `25722288cd28`

Statement

For every non-constant Boolean function f: {0,1}^k → {0,1} with k ≤ 4, let B(f) be the 2^{k-1} × 2^{k-1} biadjacency matrix of the sensitive boundary subgraph of the Hamming cube Q_k between even-parity and odd-parity vertices, defined by B(f)_{x,y} = 1 iff |x ⊕ y| = 1 and f(x) ≠ f(y). Conjecture: log_2(perm(B(f)) + 1) ≤ k · Q^dt(f), where perm denotes the matrix permanent (counting perfect matchings of the sensitive boundary). By the GPW indexing lift this gives D^cc(f ∘ IND_b) ≥ Ω(log b · k^{-1} · log_2(perm(B(f)) + 1)), a permanent-based lower bound on lifted CC.

Rationale

Sensitive edges form a bipartite subgraph of Q_k whose perfect-matching count is a global, non-spectral combinatorial invariant orthogonal to Fourier / influence-based measures already used in lifting arguments. Decision-tree depth controls the resolution of sensitive transitions: shallow trees can only sustain sensitive-edge structures supporting a limited number of disjoint perfect matchings, whereas parity-like deep functions saturate the bound. Because permanents are not spectral functionals, this route does not invoke the natural-proofs / algebrization barriers that block direct rank-style lifts.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1810.08668v2] Parity Decision Tree Complexity is Greater Than Granularity - [2209.08042v1] Decision Tree Complexity versus Block Sensitivity and Degree - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e8dfed8b.py", line 128, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e8dfed8b.py", line 103, in run_trial
    B = sensitive_boundary_matrix(f)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e8dfed8b.py", line 63, in sensitive_bou
      if  sum(int(a)  !=  int(b)  for  a,  b  in  zip(bin(f"{x:0{k-1}b}"),  bin(f"{y:0{k-
1}b}"))) == 1 and f[x] != f[y]:
                                                  ^^^^^^^^^^^^^^^^^^^
TypeError: 'str' object cannot be interpreted as an integer
```

Judge reasoning

The test crashed with a TypeError in sensitive_boundary_matrix (misuse of bin() on a formatted string) before producing any data, so the pre-registered support condition cannot be evaluated. | next: Fix the bit-comparison bug (e.g., use Hamming distance via bin(x ^ y).count('1') on integers x,y in range(2^k)) and re-run the full 2^16 enumeration on k=4.

Subdeterminant Dispersion Lower-Bounds Singular-Tail Matrix Rigidity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy theory via Chazelle's subdeterminant lower bound for hereditary discrepancy (Chazelle-Lovász-Vu / Beck-Fiala lineage) × Matrix rigidity (Valiant) measured by the singular-value tail-rank of n×n ±1 sign matrices, the standard pre-Razborov proxy for unsaturated explicit-rigidity bounds
- **Recorded**: 2026-04-27 11:13 UTC
- **Entry ID**: `09da8624d6b7`

Statement

For every n×n ±1 sign matrix M with n ≥ 4, define the k-subdeterminant dispersion Δ(M) = max over 2 ≤ k ≤ 6 of max_{S,T∈[n], |S|=|T|=k} |det(M[S,T])|^{1/k} / √k, and the energy-tail rank τ(M) = #{i : σ_i(M)² ≥ ‖M‖_F² / n}. Then τ(M) · Δ(M)² ≥ (1/4)·n. Equivalently, any matrix whose every small subdeterminant is small (low Chazelle dispersion) must have its energy spread across many singular values.

Rationale

Chazelle's subdeterminant bound is the sharpest combinatorial witness for hereditary discrepancy yet has never been linked directly to singular-tail rigidity, even though both quantify how 'flat' a sign matrix can be made by submatrix selection. Bourgain's Λ(p) heuristics suggest small subdeterminants force flat singular spectra (Kashin-style mass-spreading), which would forbid the low-rank-plus-sparse decomposition that defines rigidity collapse. A constant in the bound, if true, gives a poly-time computable explicit-rigidity lower bound that bypasses the natural-proofs barrier (Δ is not a large pseudorandom-closed property: Hadamard matrices saturate it).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_50c2f957.py", line 86, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_50c2f957.py", line 62, in run_trial
    delta = subdeterminant_dispersion(M)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_50c2f957.py", line 50, in subdeterminan
    det_val = abs(determinant(gaussian_elimination(submat)))
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_50c2f957.py", line 21, in gaussian_elim
    A[i], A[max_row] = A[max_row], A[i]
                       ~^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with an IndexError in the gaussian_elimination routine before any R values were produced, so neither the support nor the falsification criterion can be evaluated. | next: Fix the subdeterminant computation (e.g., use numpy.linalg.det or guard the pivot search) and rerun the pre-registered 1050-matrix sweep.

Lyndon Factor Count of Truth Table Lower-Bounds DT Leaves Under IND-Lifting

- **Verdict**: INCONCLUSIVE
- **Bridge**: Combinatorics on words: the Chen-Fox-Lyndon factorization of a binary word into a non-increasing product of Lyndon words (Duval's linear-time algorithm; free Lie algebra Hall basis, Lothaire ch. 5) × Decision-tree leaf count of Boolean functions, lifted via the Raz-McKenzie / Göös-Pitassi-Watson Indexing gadget to deterministic communication protocol leaf count (standard DT-to-CC lifting)
- **Recorded**: 2026-04-27 11:46 UTC
- **Entry ID**: a95d970248c4

Statement

Let f:{0,1}^n->{0,1} have truth table read in lex order as a binary word w(f) of length 2^n, and let L(f) be the number of factors in the unique Chen-Fox-Lyndon (Lyndon) factorization of w(f). Then the minimum leaf count Leaves(f) of any deterministic decision tree computing f satisfies Leaves(f) >= L(f); consequently, by Göös-Pitassi-Watson lifting, the leaf count of every deterministic protocol computing f o IND_b is at least L(f)^{Omega(b)} once b >= ceil(log2 n).

Rationale

A Lyndon factorization isolates the maximal aperiodic 'incompressible' segments of w(f); intuitively, each new Lyndon factor witnesses a position where the truth table jumps off any prior periodicity, forcing any DT to commit a distinct leaf-region to it. Combinatorics on words is essentially absent from DT/communication lower bounds, yet Duval's factorization plugs cleanly into the lifting black box (leaf count amplifies to protocol size) without invoking algebrization-sensitive structure.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1810.08668v2] Parity Decision Tree Complexity is Greater Than Granularity - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [2003.09703v1] Variance function of boolean additive convolution - [1811.10735v4] Automatic Induction of Neural Network Decision Tree Algorithms - [0906.0693v3] An improved lower bound on the counterfeit coins problem

Empirical Test

- exit code: `1`, elapsed: `130.09s`

```
TRIAL: {'metric_name': 'min_L_f', 'metric_value': 1, 'instances_tested': 65808, 'conjecture_holds':
TRIAL: {'metric_name': 'min_L_f', 'metric_value': 1, 'instances_tested': 65808, 'conjecture_holds':
TRIAL: {'metric_name': 'min_L_f', 'metric_value': 1, 'instances_tested': 65808, 'conjecture_holds':
TRIAL: {'metric_name': 'min_L_f', 'metric_value': 1, 'instances_tested': 65808, 'conjecture_holds':
TRIAL: {'metric_name': 'min_L_f', 'metric_value': 1, 'instances_tested': 65808, 'conjecture_holds':

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0487d516.py", line 92, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0487d516.py", line 92, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                           ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

The test crashed with a KeyError on 'seed' before producing valid pass/fail data; the trial outputs report 'mapping_undefined' counterexamples and conjecture_holds=False, but no genuine f with Leaves(f)<L(f) was exhibited. The pre-registered support condition was not unambiguously met, and the critic challenged the run. | next: Fix the harness so each trial result includes a 'seed' field and emits an explicit (f, Leaves(f), L(f)) triple, then re-run the full enumeration over n in {2,3,4} to obta

---

Kashin-Split Energy Gap Lower-Bounds Sign-Matrix Rigidity at Rank n/4

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrepancy / dispersion via Kashin-style orthogonal decompositions (Kashin's theorem on splitting l_2^{2n} into nearly l_1-equivalent halves; Bourgain-Tzafriri restricted invertibility flavor) × Matrix rigidity (Valiant) of n x n {+1,-1} sign matrices at target rank r = n/4, measured as the minimum Hamming weight of a perturbation E such that rank(M+E) <= r
- **Recorded**: 2026-04-27 12:12 UTC
- **Entry ID**: `223537333300`

Statement

For every n x n {+1,-1} matrix M, partition the n columns greedily into two halves (A,B) maximizing the Frobenius energy gap g(M) = | ||M_A||_F^2 - ||M_B||*F^2 | / n^2, and let s_low(M) be the sum of the bottom 3n/4 singular values of M divided by n^{3/2}. Then the rigidity at rank n/4 satisfies R_M(n/4) >= c * n^2 * (s_low(M) - g(M))*+ for an absolute constant c > 0 (we conjecture c = 1/16).

Equivalently: whenever a sign matrix has a small Kashin energy-gap (g small) but a large bottom-tail singular sum, it must be rigid.

Rationale

Kashin splittings expose the obstruction that prevents low-rank approximation from cheaply absorbing many sign-flips: when no column-bipartition concentrates Frobenius energy, the residual after removing the top n/4 spectrum cannot be packed into few sparse perturbations without violating a discrepancy-style dispersion inequality of Bourgain-Tzafriri type. This bridges the dispersion side of discrepancy theory (energy distribution under random orthogonal partitions) with Valiant's rigidity, an angle not pursued in the existing rigidity literature which focuses on either spectral lower bounds (Lokam) or paving (Alon-Panigrahy-Yekhanin).

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [0610498v4] Bounds on changes in Ritz values for a perturbed invariant subspace of a Hermitian matrix - [1005.0979v1] Supersymmetry in Random Matrix Theory - [1204.0062v4] Improved matrix algorithms via the Subsampled Randomized Hadamard Transform - [2208.05223v1] Modular Bourgain-Tzafriri Restricted Invertibility Conjectures and Johnson-Lindenstrauss Flattening Conjecture - [1503.07054v1] A Geometric Perspective on the Singular Value Decomposition

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfbe7700.py", line 121, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfbe7700.py", line 100, in run_trial
    s_low_M = s_low(M)
              ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfbe7700.py", line 93, in s_low
    _, S, _ = singular_values(M)
              ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfbe7700.py", line 36, in singular_valu
    norm_q = frobenius_norm(q)
             ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfbe7700.py", line 22, in frobenius_nor
    return sum(sum(x**2 for x in row) for row in M)**0.5
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfbe7700.py", line 22, in <genexpr>
    return sum(sum(x**2 for x in row) for row in M)**0.5
                  ^^^^^^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in frobenius_norm (it was passed a 1D vector instead of a 2D matrix during singular value computation), so no data was produced and the pre-registered support condition could not be evaluated. | next: Fix the singular_values/frobenius_norm helpers to handle 1D vectors (or use numpy.linalg.svd and numpy.linalg.norm) and re-run the full sweep across n in {6,8,10,12,14,16}.

Roe-trace pairing for Inner-Product realizes coarse depth κ ≥ log(R) and certifies a quadratic-log gap against random f

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) — coarse geometry / Roe-algebraic K-theory applied to circuit complexity (mathematical branch: noncommutative geometry & geometric group theory) × Boolean formula depth (NC^1 vs. P): explicit super-logarithmic depth lower bounds via index pairings on the KW-metric space
- **Recorded**: 2026-04-27 12:41 UTC
- **Entry ID**: `738bb754804a`

Statement

Let f = IP_n : {0,1}^{2n} → {0,1} be the inner-product-mod-2 function and let (X_f, d_f) be its KW-metric space with d_f(x,y) := log_2(min number of coordinates needed to distinguish x,y). Let c_IP ∈ HX^1(X_f) be the coarse cocycle obtained from the canonical 'parity-of-disagreement' 1-cochain c_IP(x,y) = ⟨x⊕y, x'⊕y'⟩ mod 2 (lifted to Z via signed indicator), and for each propagation parameter R ≥ 1 let T_R ∈ C_u^*[X_f] be the banded Roe operator T_R(x,y) := 1[d_f(x,y) ≤ R]/(1+|B_R(x)|). Then (i) (axiom A5 instance) the trace pairing satisfies log|⟨c_IP, T_R⟩| / log R → α ≥ 1 as n → ∞ and R = o(n), so κ(IP_n) ≥ 1 with this normalization; (ii) (axiom A3 instance) for a uniformly random f' on the same X_{f'} the same construction yields log|⟨c_IP, T_R⟩|/log R → 0 in probability, separating structured and random f.

Rationale

This is a direct empirical probe of axioms A5 (Roe-index rigidity: explicit candidates realize a nontrivial pairing whose growth rate is the coarse depth κ) and A3 (anti-natural-proofs: random instances yield trivial pairings). IP_n is the smallest 'KRW-friendly' explicit function whose KW protocol has a clean parity structure that lifts to a coarse cocycle, so we expect the pairing's logarithmic-growth exponent to be a faithful proxy for κ. Sub-additivity (A2) and Coarse-KW link (A1) both predict κ(IP_n) ≥ 1 in this normalization (a sub-claim of the conjectured Ω(log^2 n) for Andreev), making a positive numerical exponent a falsifiable prediction. A null result would either refute A5 for IP or indicate the normalization choice for d_f is too coarse — both informative.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1404.7443v2] Depth Lower Bounds against Circuits with Sparse Orientation - [1011.5800v2] Operator-valued pseudo-differential operators and the twisted index pairing - [0710.5926v2] Mod 2 cohomology of 2-local finite groups of low rank

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d19f60d5.py", line 60, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d19f60d5.py", line 34, in run_trial
    d_f = [[math.log2(hamming_weight(xor(x, y))) for y in X_f] for x in X_f]
```

```
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: math domain error
```

Judge reasoning

The test crashed with a math domain error (log2 of zero when x == y) before producing any α(n) or α’(n) data, so neither the support nor falsification conditions can be evaluated. | next: Guard the d_f computation against the diagonal (e.g., set d_f(x,x) := 0 or skip x==y pairs) and re-run for n∈{3,4,5,6} with ≥20 random-f’ seeds.

Cubical Betti Sum of Implicant Complex Lower-Bounds DNF-MCSP

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Cubical algebraic topology — total Betti number B(f)=Σ_k β_k(C_f;GF(2)) of the cubical subcomplex C_f of [0,1]^n whose k-cells are exactly the k-dimensional sub-cubes of the Boolean hypercube whose 2^k vertices all lie in supp(f); homology computed via GF(2) Smith Normal Form on the explicit cubical boundary operator (rarely deployed in circuit complexity; distinct from simplicial/persistent TDA used in data analysis) × DNF-MCSP: the meta-complexity problem of minimizing the number of product terms DNF_min(f) in a DNF representation of a Boolean function given its 2^n-bit truth table (Hirahara–Oliveira–Santhanam style restricted-MCSP variant)
- **Recorded**: 2026-04-27 12:47 UTC
- **Entry ID**: `420946ef183b`

Statement

For every non-constant Boolean function f:{0,1}^n→{0,1}, let C_f be the cubical implicant complex (k-cells = sub-cubes wholly contained in supp(f)) and let B(f)=Σ_{k=0}^{n} β_k(C_f;GF(2)). Conjecture: log_2(DNF_min(f)) ≤ B(f) for every non-constant f on n inputs, with the bound tight (up to additive 1) on parity-like functions and on disjoint-cube indicators. Equivalently, any DNF with fewer than 2^{B(f)} terms fails to compute f, providing a topological lower bound on DNF-MCSP.

Rationale

Maximal cells of C_f are precisely the prime implicants, so a DNF cover is essentially a 0-cell-cover witnessing nerve-lemma collapse of C_f; any topological obstruction (β_0 components, β_1 cycles bridging clusters, higher voids) forces additional terms beyond the naive component count. While β_0 alone has been touched on tangentially, the *full* GF(2) cubical Betti sum has not been linked to DNF size because cubical (vs simplicial) homology is unfashionable and rarely computed on truth tables. If true, cubical homology yields a barrier-respecting lower-bound technique on DNF-MCSP and a fresh handle on Hirahara-style hardness magnification.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1501.01331v1] DNF complexity of complete boolean functions - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [2003.09703v1] Variance function of boolean additive convolution - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2110.02458v3] Magnitude homology of graphs and discrete Morse theory on Asao-Izumihara complexes

Empirical Test

- exit code: `0`, elapsed: `0.25s`

```
TRIAL: {'metric_name': 'Betti Sum - DNF Min', 'metric_value': 0, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'Betti Sum - DNF Min', 'metric_value': 0, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'Betti Sum - DNF Min', 'metric_value': 0, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'Betti Sum - DNF Min', 'metric_value': 0, 'instances_tested': 0, 'conjecture_ho
TRIAL: {'metric_name': 'Betti Sum - DNF Min', 'metric_value': 0, 'instances_tested': 0, 'conjecture_ho
RESULT: FALSIFIED counterexample="singular_matrix" first_failing_seed=11
```

Judge reasoning

The test reported FALSIFIED with counterexample='singular_matrix' but instances_tested=0 across all trials, indicating a Betti-computation implementation failure (singular GF(2) matrix) rather than a genuine functional counterexample. No actual f was evaluated, so neither support nor falsification of the conjecture is demonstrated. | next: Fix the GF(2) boundary-rank computation (e.g., use a robust Smith normal form or sparse rank routine) so B(f) can be evaluated on real instances, then re-run

---

Zeckendorf Length of NW Design Rows Bounds PRG Parity Bias

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Zeckendorf / Ostrowski numeration: representation of integers as sums of non-consecutive Fibonacci numbers and the resulting digit-length statistic z(N) (Lekkerkerker–Carlitz metric theory of Fibonacci expansions, rarely deployed outside number theory) × Output bias of the Nisan–Wigderson pseudorandom generator NW_D^f against parity (linear χ_T) distinguishers, where D is a combinatorial (d,m,k,t)-design and f is a Boolean hard function
- **Recorded**: 2026-04-27 13:16 UTC
- **Entry ID**: `b133863cfef2`

Statement

Let D = {S_1,…,S_m} be any (d,m,k,t)-NW design on [d], and let z_i be the number of Fibonacci summands in the Zeckendorf representation of N_i = Σ_{j∈S_i} 2^{j-1}; set Z(D) = (1/m) Σ_i z_i. Then for every Boolean f:{0,1}^k→{0,1} whose largest non-empty Fourier coefficient max_{∅≠U⊆[k]} |f̂(U)| equals η, the NW-PRG output bias against any non-empty parity test obeys |bias_T(D,f)| := |2^{-d} Σ_{s∈{0,1}$^{d\}(-1)}${⊕*{i∈T} f(s*|{S_i})}| ≤ η^{|T|} · 2^{t·(|T| choose 2)} · (1 + Z(D)/k)^{-1}. A single (D,f,T) with k≤5, m≤8, d≤10 violating this inequality refutes the conjecture.

Rationale

NW analyses bound bias by η$^{\{|T|\}\cdot 2}${t·C(|T|,2)} (the standard Nisan–Wigderson lemma); the new factor (1+Z(D)/k)^{-1} predicts that designs whose row-indicators are arithmetically 'simple' in the Fibonacci base (low Zeckendorf length, i.e., concentrated bit patterns) leak more correlation through f, while rows with maximally spread Zeckendorf decompositions act more like Beatty-equidistributed indices and force tighter cancellation. This bridges Zeckendorf metric theory with NW pseudorandomness — a pairing with effectively no prior literature — and offers a falsifier-friendly tightening of the classical bound that is meaningful only when Z(D) is non-trivially correlated with the design's discrepancy.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1311.6531v3] Brains and pseudorandom generators - [2404.12647v1] Simple constructions of linear-depth t-designs and pseudorandom unitaries - [2601.03730v1] Perception-Aware Bias Detection for Query Suggestions - [1906.10962v3] The Fibonacci Sequence and Schreier-Zeckendorf Sets - [1302.6583v1] A note on closed-form representation of Fibonacci numbers using Fibonacci trees

Empirical Test

- exit code: `1`, elapsed: `0.30s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6b011f3.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d6b011f3.py", line 95, in run_trial
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)
                       ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError before any trial data was collected, so neither the support nor falsification criterion was evaluated. | next: Fix the trial generation so designs are actually produced (ensure results list is non-empty), then rerun the pre-registered sweep over k∈{3,4,5}, m∈{4..8}, t∈{1,2}.

Sturmian Factor-Complexity Lower-Bounds DNF-MCSP Term Count

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symbolic dynamics on the Boolean hypercube: subword/factor complexity p_w(k) of the Gray-code linearization of a truth table, in the Morse-Hedlund / Sturmian tradition (rarely deployed in meta-complexity) × DNF-MCSP: minimum number of product terms DNF_min(f) needed to represent a Boolean function f:{0,1}^n→{0,1} given its 2^n-bit truth table (Hirahara–Oliveira–Santhanam restricted-MCSP variant)
- **Recorded**: 2026-04-27 13:49 UTC
- **Entry ID**: `848aa89a6626`

Statement

Let w(f) be the binary word of length 2^n obtained by listing f's truth table in standard reflected Gray-code order, and let P(f)=max_{1≤k≤n} p_{w(f)}(k)/k where p_w(k) is the number of distinct length-k factors of w. Then DNF_min(f) ≥ ⌈P(f)/2⌉ for every non-constant f on n≤6 variables, with equality achieved by every parity function χ_S restricted to its essential variables. Equivalently: any f whose Gray-code word realizes factor-complexity ratio P(f)>2T cannot be expressed as a DNF with ≤T terms.

Rationale

Gray-code traversal makes adjacent truth-table positions differ in exactly one input bit, so a single AND-term contributes a contiguous arithmetic-progression-like block to w(f); bounding the number of distinct factors therefore bounds the combinatorial diversity of term-boundaries, which in turn bounds DNF_min. Symbolic dynamics has been heavily used in automata theory and formal languages but, to our knowledge, never linked to MCSP-style term-minimization; Sturmian/Morse-Hedlund growth gives a clean, computable scalar invariant. Because the bound is a simple counting inequality on a 2^n-bit word it sidesteps relativization (the mapping is purely syntactic on the truth table).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1303.5057v1] GRay: a Massively Parallel GPU-Based Code for Ray Tracing in Relativistic Spacetimes - [1203.2323v2] The greedy flip tree of a subword complex - [2412.05161v1] DNF: Unconditional 4D Generation with Dictionary-based Neural Fields - [1501.01331v1] DNF complexity of complete boolean functions - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube

Empirical Test

- exit code: `124`, elapsed: `180.08s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s without producing any RESULT line, so the pre-registered support condition cannot be evaluated and the critic's challenge stands. | next: Optimize the DNF_min computation (e.g., use Quine-McCluskey with caching or an ILP solver) and reduce sample sizes or parallelize so the test completes within the timeout.

Haar-Wavelet Star-Discrepancy of Sign Matrix Lower-Bounds Rigidity at Rank n/2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy via 2D Haar-wavelet star-discrepancy of ±1 sign matrices (Roth-Schmidt / Matousek L^2 anchored-box discrepancy on the dyadic grid; rarely deployed against rigidity) × Matrix rigidity (Valiant) at target rank r = n/2: minimum Hamming weight R(M,n/2) of a perturbation E with rank(M+E) ≤ n/2 over the rationals, the standard pre-Razborov rigidity proxy
- **Recorded**: 2026-04-27 14:18 UTC
- **Entry ID**: `bb503edce1c2`

Statement

For every n×n sign matrix M with n a power of 2, let D(M) = max over dyadic anchored boxes B ⊆ [n]×[n] of |Σ_{(i,j)∈B} M_{ij}| / sqrt(|B|), the normalized Haar star-discrepancy. We conjecture R(M, n/2) ≥ c · D(M)^2 / log n for some absolute c > 0, and in particular for the Hadamard sign matrix H_n we conjecture D(H_n) = Θ(sqrt(n)) so that R(H_n, n/2) = Ω(n / log n). One n ≤ 16 sign matrix M with rank(M+E) ≤ n/2 achievable at Hamming weight < c·D(M)^2/log n falsifies it.

Rationale

Star-discrepancy on the dyadic anchored-box system measures how badly a sign pattern aligns with low-rank Haar tensor projectors, and any low-rank perturbation can only destroy O(rank·log n) Haar coefficients; thus a large worst-anchored-box imbalance should force many sign flips to reach rank n/2. This bridges Roth-Schmidt L^2 discrepancy machinery — never directly fed into Valiant rigidity — to the explicit-Hadamard frontier where existing rank-trick bounds saturate at Ω(n^2/r · log(n/r)). The constructive mapping (enumerate dyadic boxes, sum entries, normalize) is computable in O(n^2 log^2 n) and gives a falsifier-friendly quantitative invariant.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [0702067v1] The Haar Wavelet Transform of a Dendrogram: Additional Notes - [1005.0979v1] Supersymmetry in Random Matrix Theory - [1112.1588v2] Evaluating Matrix Functions by Resummations on Graphs: the Method of Path-Sums - [0909.2030v2] Size Bounds for Conjunctive Queries with General Functional Dependencies - [1308.3946v1] Optimal Algorithms for Testing Closeness of Discrete Distributions

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ec3b34.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ec3b34.py", line 86, in run_trial
    H_n = hadamard_matrix(n)
          ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ec3b34.py", line 35, in hadamard_matr
    H = hadamard_matrix(n // 2)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_63ec3b34.py", line 39, in hadamard_matr
    bottom_right = [-x for x in H]
                    ^^
TypeError: bad operand type for unary -: 'list'
```

Judge reasoning

The test crashed with a TypeError in hadamard_matrix (negating a list of lists instead of elements), so no instances were evaluated and neither the support nor falsification condition can be assessed. | next: Fix hadamard_matrix to recurse on numpy arrays (or negate element-wise) and rerun the full 5-seed × ≥200-instance protocol for n ∈ {4,8,16}.

Roe-cohomological concentration: HX^1(X_f) vanishes for random f, separating natural-proof properties from κ

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); coarse geometry / Roe algebras over finite KW-metric spaces × Boolean formula-depth lower bounds (NC^1 vs. P) — a complexity-theoretic invariant on the family of n-variate boolean functions
- **Recorded**: 2026-04-27 14:43 UTC
- **Entry ID**: 7c5ad65991c5

Statement

Let f : {0,1}^n -> {0,1} be drawn uniformly at random and let (X_f, d_f) be the KW-metric space with d_f instantiated as Hamming distance on X_f = f^{-1}(0) ⊔ f^{-1}(1). For the controlled radius R = 1 and the discrete coarse coboundary operator δ on the Roe-style cochain complex C^*_{coarse}(X_f), define h^1(f) := dim_R HX^1(X_f) = dim ker(δ^1) − dim im(δ^0). The sub-conjecture (instance of axiom A3, anti-natural-proofs) asserts: (i) for n ∈ {3,4,5,6} the empirical mean of h^1(f) over uniform f tends to 0 as n grows, with Pr[h^1(f) > 0] ≤ exp(−Ω(n)); (ii) for the explicit family f_n = PARITY_n, h^1(f_n) ≥ n − 1, so the trace pairing ⟨c, T⟩ realizes κ(PARITY_n)

≥ log(R)·(n−1)/n with R = 2; (iii) consequently κ is not a 'large' property in the Razborov–Rudich sense, since it fails the largeness criterion on the uniform ensemble while remaining nonzero on explicit functions, blocking the natural-proofs obstruction to A1.

Rationale

This directly tests axiom A3 (Roe-algebraic concentration of measure). A3 predicts that κ-based proofs evade Razborov–Rudich because random f have trivial coarse cohomology — the d_f-metric of a generic f is too 'uniformly mixed' to support a nontrivial Roe-controlled cocycle, while structured f (PARITY, AND-OR towers) inherit nontrivial HX^1 from their algebraic stratification of {0,1}^n. If concentration fails (random f routinely have h^1 > 0), then A3 is wrong and the framework collides with the natural-proofs barrier; if it holds, κ is a 'small' invariant that still distinguishes explicit functions, which is exactly the regime needed for A1 to give a lawful P ⊄ NC^1 attack.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [0805.1385v3] Almost-natural proofs - [2003.09703v1] Variance function of boolean additive convolution

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb2e6197.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb2e6197.py", line 40, in run_trial
    G = {i: [j for j in range(len(X)) if hamming_distance(i, j) <= 1] for i in range(len(X))}
                                         ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb2e6197.py", line 18, in hamming_dista
    return sum(xi != yi for xi, yi in zip(x, y))
                                      ^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) inside hamming_distance before producing any data, so no empirical results were collected for either the random ensemble or PARITY_n. | next: Fix the bug by representing X elements as bit-vectors (e.g., tuples or binary strings) rather than ints, or convert i,j to their binary representations inside hamming_distance, then rerun the trials.

Antichain-Width of Query Certificate Poset Bounds Lifted IND CC

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Order theory / Dilworth's theorem on the poset of minimal 0/1-certificates of a Boolean function ordered by sub-assignment, with antichain-width w(f) computed via Koenig-style maximum matching on the certificate bipartite graph (rarely deployed in lifting-theorem lower

bounds; distinct from rank/Mobius approaches) × Deterministic two-party communication complexity D^cc(f o IND_b) of Indexing-gadget-lifted Boolean functions, accessed through deterministic decision-tree depth Q^dt(f) via the Raz-McKenzie / Goos-Pitassi-Watson lifting theorem

- **Recorded**: 2026-04-27 14:50 UTC
- **Entry ID**: `52be4733a1c4`

Statement

For every Boolean function f:{0,1}^n -> {0,1} with n <= 12, let w(f) be the Dilworth antichain-width of the poset P(f) whose elements are the minimal 0-certificates and minimal 1-certificates of f, ordered by sub-assignment refinement (a <= b iff dom(a) subset of dom(b) and a agrees with b on dom(a)). Then Q^dt(f) >= ceil(log_2(w(f) + 1)), and consequently the IND_b-lifted deterministic communication complexity satisfies D^cc(f o IND_b) >= ceil(log_2(w(f)+1)) * log_2(b) - O(log n). Falsifier: a single f with w(f) >= $2^{(Q}$dt(f)+1).

Rationale

Lifting theorems convert decision-tree depth to communication depth, but the depth side is usually bounded via adversary or rank arguments; Dilworth width on the certificate poset captures a combinatorial ‘spread’ of incompatible partial proofs that any decision tree must distinguish, giving a poset-theoretic depth bound that should compose cleanly under IND-lifting. Because antichains of minimal certificates are pairwise incompatible, every deterministic tree must reach distinct leaves on each, forcing log_2(w) depth without invoking algebrization-sensitive algebra. If true, the bound provides a purely order-theoretic certificate that survives the gadget product, opening a non-rank route to monotone-style depth separations.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [2211.17214v1] Lifting to Parity Decision Trees Via Stifling - [1302.4207v1] A composition theorem for decision tree complexity - [1306.1593v2] The root posets and their rich antichains - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube

Empirical Test

- exit code: `1`, elapsed: `0.39s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2595fd2f.py", line 137, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2595fd2f.py", line 118, in run_trial
    w_f = maximum_bipartite_matching(poset)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2595fd2f.py", line 69, in maximum_bipar
    dfs(u)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2595fd2f.py", line 59, in dfs
    for v in graph[u]:
             ~~~~~^^^
KeyError: 0
```

Judge reasoning

The test crashed with a KeyError in the bipartite matching routine before producing any data, so neither the support nor falsification conditions can be evaluated. | next: Fix the maximum_bipartite_matching function to initialize graph[u] as an empty adjacency list (e.g., via defaultdict(list)) and rerun the pre-registered protocol.

Apolar Span of Clause-Product Polynomial Bounds DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Apolarity / partial-derivative-span (Macaulay inverse system, catalecticant rank) — the Iarrobino–Kanev / Landsberg–Ottaviani lineage that underlies GCT border-rank obstructions for permanent vs. determinant. × Lexicographic DPLL search-tree leaf count on small unsatisfiable 3-CNF formulas.
- **Recorded**: 2026-04-27 15:25 UTC
- **Entry ID**: `fe0af6ae6d43`

Statement

For an unsatisfiable 3-CNF F on n ≤ 10 variables with m clauses, attach to each clause C the affine linear form L_C := off(C) + Σ_{positive x_i ∈ C} x_i − Σ_{negative ¬x_i ∈ C} x_i, where off(C) = (#negative literals in C); thus L_C(x) equals the number of true literals of C at x ∈ {0,1}^n. Let f_F := ∏*{C ∈ F} L_C ∈ Q[x_1,…,x_n], let π : Q[x] → Q[x]/(x_i^2 − x_i) □ Q^{2^n} be the squarefree-reduction map, and define μ(F) := dim_Q span{π(∂*{x_i} f_F) : i=1,…,n} ⊆ Q^{2^n}, computed via the identity π(∂*{x_i} f_F) = Σ*{C} s(C,i)·π(∏*{C′≠C} L*{C′}) where s(C,i) ∈ {−1,0,+1} is the sign of variable x_i in clause C. Conjecture: for every unsatisfiable F in this regime, μ(F) ≥ ⌈log_2 D_F⌉, where D_F is the leaf count of depth-first DPLL with unit propagation that always branches on the lowest-indexed unassigned variable (positive branch first); a single unsatisfiable F with μ(F) < ⌈log_2 D_F⌉ refutes it.

Rationale

Apolarity converts lower bounds on partial-derivative-span dimension into border-rank obstructions, the engine of Landsberg–Ottaviani's permanent lower bounds inside GCT. Each clause is genuinely a linear form, so f_F is exactly the kind of degree-m product whose derivative-span GCT analyzes for orbit-closure obstructions; the squarefree projection π is the natural finite Macaulay quotient. Linking μ(F) to DPLL leaves tests whether the algebraic rigidity that obstructs orbit-closure containment is the same rigidity that obstructs DPLL collapse — a candidate computational shadow of GCT obstructions.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [2405.16149v3] Small unsatisfiable k-CNFs with bounded literal occurrence - [1203.3706v2] On the Complexity of Computing Minimal Unsatisfiable LTL formulas

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
ome/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 159, in <module>
    result = run_trial(seed)
```

```
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 115, in run_trial
    if is_unsat(F, n):
       ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 107, in is_unsat
    return not dpll(F, assignment, n)
               ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 85, in dpll
   if dpll(F - {tuple(sorted([var + 1 if sign > 0 else -(var + 1) for sign, var in clause])) for clause i
     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 85, in dpll
   if dpll(F - {tuple(sorted([var + 1 if sign > 0 else -(var + 1) for sign, var in clause])) for clause i
     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 85, in dpll
   if dpll(F - {tuple(sorted([var + 1 if sign > 0 else -(var + 1) for sign, var in clause])) for clause i
     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 993 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38d58c5e.py", line 80, in dpll
    if all(lit in assignment and assignment[lit] == sign for lit, sign in clause):
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

The test crashed with a RecursionError in the DPLL implementation before producing any data, so no instances were evaluated against the pre-registered criterion. | next: Fix the DPLL routine (iterative implementation or raise recursion limit, and correct the buggy clause-filtering expression) and rerun the 30-instance protocol.

Cheeger Constant of Quotient Multigraph Bounds Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral graph theory via Cheeger constants of edge-contraction quotient multigraphs (h(G/S) over induced vertex partitions S, rarely applied to proof complexity) × Resolution refutation width w(T(G,c) ⊢ ⊥) of Tseitin formulas on small 3-regular graphs with odd total charge
- **Recorded**: 2026-04-27 15:55 UTC
- **Entry ID**: `d25429e913d1`

Statement

For every 3-regular graph G on n ≤ 20 vertices with odd-total charge c, let h*(G) = min over balanced bipartitions S of |E(S, V§)|/min(|S|,|V§|), and let h_q(G) = min over partitions P of V into k≥2 nonempty blocks of (cut-edges of P)/(k−1). We conjecture w(T(G,c) ⊢ ⊥) ≥ ⌈h_q(G)⌉ and moreover w(T(G,c) ⊢ ⊥) ≤ 3·h_q(G) + 2 across all such (G,c). Equivalently, the resolution width of Tseitin equals the quotient-Cheeger constant up to a constant factor of 3, sharper than the standard expansion bound which uses only h*(G).

Rationale

Classical Ben-Sasson–Wigderson and Urquhart bounds use vertex-expansion of subsets of size n/3..2n/3 to prove Ω(h*) width lower bounds, but they discard cut information from finer partitions. The quotient-multigraph Cheeger constant h_q aggregates parity obstructions across all coarsenings of G, exactly matching the inductive contraction structure used in resolution derivations of

Tseitin (each contraction merges two odd-charge components). If true, this gives a tight combinatorial characterization—not just a lower bound—of Tseitin width, sidestepping natural-proofs (it is non-uniform and depends on the specific G), relativization, and algebrization.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1807.02225v2] Cheeger inequalities for graph limits - [1411.3530v4] Cheeger constants, structural balance, and spectral clustering analysis for signed graphs

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fcf62212.py", line 136, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fcf62212.py", line 108, in run_trial
    G = generate_3_regular_graph(n)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fcf62212.py", line 35, in generate_3_re
    raise ValueError("Invalid number of vertices for a 3-regular graph")
ValueError: Invalid number of vertices for a 3-regular graph
```

Judge reasoning

The test crashed with a ValueError in generate_3_regular_graph before producing any data, so neither the lower nor upper bound was evaluated on any (G,c) pair. No RESULT line was produced and the pre-registered criterion of ≥30 pairs was not met. | next: Fix generate_3_regular_graph to handle valid even n∈{6,8,10} (3-regular graphs require even n), then rerun with exact h_q enumeration and width-bounded BFS prover across ≥30 (G,c) pairs.

RSK Second-Row Length of f-Sorted Permutation Bounds Lifted IND CC

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Plactic monoid combinatorics via Robinson-Schensted-Knuth (RSK) insertion and Schensted's theorem on Young tableau shapes (Lothaire, Stanley EC2 Ch. 7); the longest weakly-decreasing-subsequence statistic λ_2 of a permutation, accessed by patience sorting × Deterministic decision-tree depth Q^dt(f) of small Boolean functions f:{0,1}^k→{0,1}, lifted to deterministic two-party communication complexity D^cc(f∘IND_b) via the Raz-McKenzie / Göös-Pitassi-Watson indexing-gadget lifting theorem
- **Recorded**: 2026-04-27 16:27 UTC
- **Entry ID**: `a991258bde76`

Statement

For f:{0,1}^k→{0,1}, let π_f be the permutation of [0..2^k−1] obtained by stable-sorting the inputs by the key (f(x), x), and let λ(f)=(λ_1,λ_2) be the shape of the RSK insertion tableau P(π_f). Conjecture: ⌈log_2(λ_2(f)+1)⌉ ≤ Q^dt(f) for every Boolean f, with equality achievable by parity-like functions. By GPW lifting with indexing gadget IND_b, this entails D^cc(f∘IND_b) ≥ ⌈log_2(λ_2(f)+1)⌉·log_2 b − O(log k).

Rationale

Schensted's theorem makes λ_2(π_f) the length of the longest weakly-decreasing subsequence of π_f, a global obstruction distinct from antichain width, matroid rank, or Möbius invariants — it captures how thoroughly the value-class boundary scrambles the lex order. Each decision-tree query can split the truth table into two halves but cannot shorten a long alternating descent chain by more than a factor of 2, suggesting Q^dt must grow logarithmically with λ_2. The construction is a simple permutation statistic, side-steps the natural-proofs barrier (λ_2 is not a pseudorandom property — random f has λ_2≈c·2^{k/2}), is unaffected by algebrization (no oracle-relativizing structure), and RSK has never been deployed inside the GPW lifting framework.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1808.06095v5] Skew RSK and the switching on ballot tableau pairs - [1810.08668v2] Parity Decision Tree Complexity is Greater Than Granularity - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [0109079v2] GL(n,q) and Increasing Subsequences in Nonuniform Random Permutations - [2309.01208v1] Streaming and Query Once Space Complexity of Longest Increasing Subsequence

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4eae8e14.py", line 89, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4eae8e14.py", line 63, in run_trial
    pi_f = sorted(range(n), key=lambda x: (f(x, j) for j in range(k)))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: '<' not supported between instances of 'generator' and 'generator'
```

Judge reasoning

The test crashed with a TypeError before producing any data — the sort key was incorrectly written as a generator expression rather than a tuple, so no functions were evaluated and no RESULT line was produced. | next: Fix the key in the sort to `key=lambda x: (f(x), x)` (using the precomputed truth table f[x]) and rerun the full enumeration over all 65536 functions on k=4.

Coarse-dimensional KRW direct sum: asdim is sub-additive under block composition

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework; coarse geometry / asymptotic-dimension theory (Gromov, Bell–Dranishnikov) × Boolean formula-depth complexity; the Karchmer–Raz–Wigderson (KRW) direct-sum conjecture for NC^1 lower bounds
- **Recorded**: 2026-04-27 16:44 UTC
- **Entry ID**: `f685d342f4c0`

Statement

For all Boolean functions f:{0,1}^n -> {0,1} and g:{0,1}^m -> {0,1}, let (X_f, d_f) and (X_g, d_g) denote their KW-metric spaces and let (X_{f∘g}, d_{f∘g}) be the coarse_product encoding the block composition f ∘ g. Then there exists a universal constant C ≤ 2 such that asdim(X_{f∘g}) ≥ asdim(X_f) + asdim(X_g) − C. Equivalently, certified by the asymptotic-dimension-witness primitive: if χ_f, χ_g are optimal (k_f, R_f)- and (k_g, R_g)-witnesses for X_f, X_g respectively, then no (k_f + k_g − C − 1, R)-witness exists for X_{f∘g} for any finite R. Combined with axiom A1, this yields the KRW-style implication d(f ∘ g) ≥ Ω(d(f)) + Ω(d(g)) − O(1).

Rationale

This sub-conjecture tests axiom A2 (Composition sub-additivity) directly using the framework's coarse_product primitive. It mirrors the classical Bell–Dranishnikov inequality asdim(X × Y) ≤ asdim(X) + asdim(Y) for metric products, asserting that the matching lower bound also holds for the depth-adjusted KW-product metric d_{f∘g}. The mechanism is structural: a controlled cover of X_{f∘g} of multiplicity m must, when restricted to fibers, refine into controlled covers of X_f and X_g, so the multiplicity exponents add. Verifying A2 on small instances closes the gap between κ-additivity and KRW, since Andreev-style amplification then bootstraps κ-bounds through composition (Axiom A5).

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [0711.0446v1] Asymptotic Properties of Hilbert Geometry - [2306.00615v5] Toward Better Depth Lower Bounds: A KRW-like theorem for Strong Composition - [2307.07605v1] First-order Methods for Affinely Constrained Composite Non-convex Non-smooth Problems: Lower Complexity Bound and Near-o

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2678b93a.py", line 158, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2678b93a.py", line 110, in run_trial
    d_f_val = max(d_f(f, x, y) for x, y in itertools.combinations(X_f, 2))
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2678b93a.py", line 110, in <genexpr>
    d_f_val = max(d_f(f, x, y) for x, y in itertools.combinations(X_f, 2))
                      ^
NameError: name 'f' is not defined
```

Judge reasoning

The test crashed with a NameError ('f' is not defined') before producing any data, so the pre-registered support condition could not be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the bug in run_trial (line 110) by passing f as an argument to d_f within the generator expression, then re-run the deterministic enumeration over all 36 (f,g) pairs.

Fourier Variance Ratio of Violated-Clause Counter Bounds Tree-Resolution Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Walsh-Hadamard Fourier analysis of integer-valued lattice functions on $\{0,1\}^n$ (Friedgut/Bourgain/KKL-style concentration applied to non-Boolean counters, distinct from the $\{-1,+1\}$ indicator regime) × Tree-like resolution refutation depth $d_T(F\vdash\bot)$ of small unsatisfiable 3-CNFs, equivalently the minimum height of any DPLL search-tree certifying UNSAT
- **Recorded**: 2026-04-27 16:59 UTC
- **Entry ID**: `ee44e38271ee`

Statement

Let F be an unsatisfiable 3-CNF on n variables with m clauses, and let $k_F:\{0,1\}^n\to\mathbb{Z}_{\geq 0}$ be the violated-clauses counter $k_F(x)=\#\{j: x\not\models C_j\}$, expanded as $k_F=\Sigma_S \hat{k}F(S)\chi_S$ *in the Walsh-Hadamard basis. Define the dimensionless Fourier variance ratio* $R(F)=Var(k_F)/E[k_F]^2$ *over uniform* $x\in\{0,1\}^n$ *(equivalently* $\Sigma\{S\neq\emptyset\}\hat{k}_F(S)^2$ divided by $\hat{k}_F(\emptyset)^2$). Conjecture: every UNSAT 3-CNF F with $E[k_F]\geq 1$ satisfies $d_T(F\vdash\bot) \geq \lceil \log_2(1/R(F))\rceil + 1$; a single F with $d_T(F)$ strictly below this Fourier-concentration floor falsifies the claim.

Rationale

A small variance ratio R(F) means k_F is sharply concentrated around its mean, so the Friedgut/Bourgain noise picture predicts that random restrictions of small radius almost surely preserve at least one violated clause, forcing every branch of any DPLL tree to descend at least $\Theta(\log(1/R))$ levels before some clause empties. The integer-counter Fourier spectrum captures this concentration directly through Plancherel on k_F^2, bridging KKL-style spectral concentration with the proof-complexity primitive of search-problem decision trees in a manner distinct from prior Boolean-indicator influence bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0806.1148v2] Unsatisfiable CNF Formulas need many Conflicts - [0905.1587v5] Unsatisfiable Linear CNF Formulas Are Large and Complex - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [1508.06336v1] SPRIGHT: A Fast and Robust Framework for Sparse Walsh-Hadamard Transform - [2101.09528v1] Hard satisfiable formulas for DPLL algorithms using heuristics with small memory

Empirical Test

- exit code: `124`, elapsed: `180.08s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any RESULT line or data, so neither the support nor the falsification condition can be evaluated. | next: Reduce the instance set or n-range (e.g., n∈{8,10}) and/or replace exact tree-resolution depth computation with a faster solver/bound, then rerun with a longer timeout to obtain support_fraction and Spearman ρ.

Frobenius Trace Defect of Clause-Indicator Polynomial Lower-Bounds ACC^0[2] Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Arithmetic of polynomial residues over F_2[x]/(x^p - x): the Frobenius-trace defect tr_p(P) := dim_{F_2} (P - P^p mod (x^p - x)) computed by direct evaluation, in the Lucas/Kummer combinatorial-trace tradition (rarely applied to circuit complexity) × ACC^0[2] circuit size for Boolean functions presented as a CNF formula, the Williams-Murray target class for NEXP-ACC separations
- **Recorded**: 2026-04-27 17:25 UTC
- **Entry ID**: `b1916c940fcb`

Statement

Let F be an unsatisfiable 3-CNF on n variables and let P_F(x_1,...,x_n) := sum over clauses C of prod_{l in C}(1 - l) be its clause-indicator polynomial reduced mod 2. Define the Frobenius-trace defect at prime p=3 as d_3(F) := |{a in F_3^n : P_F(a)^3 != P_F(a) (mod 3)}|. Then the minimum ACC^0[2] circuit size S(F) satisfies log_2 S(F) >= c * d_3(F) / 3^{n/2} for an absolute constant c >= 1/4 across all unsatisfiable 3-CNFs with n <= 20.

Rationale

The Williams-Murray algorithmic approach exploits that ACC circuits of size s admit succinct polynomial representations whose Frobenius-orbit structure mod small primes governs satisfiability counting; a non-vanishing trace defect mod 3 is exactly the obstruction that prevents collapse of P_F into a small-degree symmetric form and therefore lower-bounds any ACC^0[2] simulation. Computing d_3(F) by direct enumeration over F_3^n is feasible for n <= 20 (~3^20 ≈ 3.5e9 — restrict to n<=12 in tests), and S(F) is upper-bounded by enumerating small {AND,OR,NOT,MOD2} circuits, giving a falsifier-friendly inequality. The bridge is rare because Frobenius-trace combinatorics on clause polynomials has not, to our knowledge, been tied to ACC simulation size.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2311.04204v3] Sharp Thresholds Imply Circuit Lower Bounds: from random 2-SAT to Planted Clique - [2603.27466v1] English translation of Frobenius' and Stickelberger's "On the theory of elliptic functions" - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9450f5e2.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9450f5e2.py", line 88, in run_trial
    d_3 = frobenius_trace_defect(poly, n)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9450f5e2.py", line 45, in frobenius_tra
   if evaluate_polynomial(poly, assignment) ** 3 != evaluate_polynomial(poly, assignment):
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9450f5e2.py", line 32, in evaluate_poly
    if assignment[var] == 0:
```

```
      ~~~~~~~~~~^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so the pre-registered support condition could not be evaluated. Per rule 2, a crash mandates an INCONCLUSIVE verdict. | next: Fix the variable-indexing bug in evaluate_polynomial (likely an off-by-one between 1-indexed clause literals and 0-indexed assignment list) and rerun the full 200-instance protocol across n ∈ {6,8,10,12}.

Oracle Collapse of κ: Anti-Relativization Stress Test for the Coarse-KW Invariant (A4)

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework; underlying math: coarse geometry / Roe C-*algebras and metric geometry of boolean functions, with Mendel–Naor style metric-invariant obstructions × Relativized boolean formula depth / NC^1 vs P^A separation barriers — specifically, the complexity-theoretic object 'oracle-augmented formula depth d^A(f) over the basis B ∪ {f-gate}', and the induced uniform Roe algebra C_u*^[X_{f^A}] viewed as a complexity invariant
- **Recorded**: 2026-04-27 17:34 UTC
- **Entry ID**: `a81b4af0bc2b`

Statement

Let f:{0,1}^n → {0,1} be any boolean function, and let f^A denote the oracle-augmented variant whose KW-metric d_{f^A} is built from depth-bounded distinguishers that may use a single oracle gate computing f. Then (i) d_{f^A}(x,y) ≤ 1 for every (x,y) ∈ f^{-1}(0) × f^{-1}(1); (ii) the resulting controlled cover U has propagation R ≤ 1, so the algebraic uniform Roe algebra C_u^*[X_{f^A}] is the full matrix algebra on X_{f^A}; (iii) $HX^{1(X_{f}}$A}) = 0 because every coarse 1-cochain is a coboundary on a diameter-1 space; hence (iv) the trace pairing ⟨c,T⟩ vanishes for all admissible (c,T), forcing κ(f^A) = O(1) uniformly in n, while κ(f) can be ω(1). This realizes axiom A4: the coarse depth κ is destroyed by oracle access and therefore cannot relativize.

Rationale

This isolates and tests axiom A4 (anti-relativization). The conjecture should hold because the KW-metric is *defined* via depth-bounded distinguishers; an oracle gate for f trivially separates f^{-1}(0) from f^{-1}(1) in one step, collapsing all nontrivial distances to 1. Once the metric collapses to a 1-bounded space, the Roe cohomology HX^1 becomes a finite-dimensional coboundary space and the index pairing tr(c·T) is forced to zero, so κ degenerates. Confirming this empirically validates that κ is a genuine metric invariant (not algebraic), distinguishing it from relativizing measures like communication complexity, and aligning with Mendel–Naor metric-cotype obstructions (A4) and Yu-style coarse rigidity (A5, used here only as the trace formalism).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2112.03677v7] On Baker-Gill-Solovay Oracle Turing Machines and Relativization Barrier - [2601.22054v1] MetricAnything: Scaling Metric Depth Pretraining with Noisy Heterogeneous Sources

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e4b213a.py", line 145, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e4b213a.py", line 38, in run_trial
    X_f = [(i, j) for i in range(2**n) for j in range(2**n) if f(i) != f(j)]
                                                               ^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e4b213a.py", line 22, in AND_3
    return x[0] and x[1] and x[2]
           ~^^^
TypeError: 'int' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError (treating an int as subscriptable in AND_3) before producing any data, so none of the pre-registered support conditions could be evaluated. | next: Fix the boolean function input handling (decode the integer i into a bit-tuple via something like tuple((i»k)&1 for k in range(n)) before passing to f) and rerun the protocol.

Carries-Polynomial Degree Lower-Bounds EF-Lines for PHP_n

- **Verdict**: INCONCLUSIVE
- **Bridge**: p-adic carry combinatorics: Kummer's theorem on the 2-adic valuation of binomial coefficients via base-2 carry counts, lifted to the 'carries polynomial' C_n(x,y) = sum over (i,j) of [carry occurs at bit position k when adding i and j in binary] x^i y^j; an elementary number-theoretic gadget rarely deployed in proof complexity × Extended Frege (EF) proof complexity: number of distinct lines (sub-formulas after eliminating extension axioms) in a tree-like EF refutation of the pigeonhole principle PHP_n^{n+1} encoded as a CNF
- **Recorded**: 2026-04-27 17:36 UTC
- **Entry ID**: 151708c9a4ad

Statement

Let kappa(n) be the GF(2)-rank of the n x n matrix M_n with M_n[i,j] = parity of the number of binary carries when adding i and j (0<=i,j<n). The conjecture: for every tree-like EF refutation Pi of PHP_n^{n+1}, the number of distinct sub-formulas appearing in Pi (the 'EF line count') is at least kappa(n) + n, and asymptotically kappa(n) = n - O(log n) so this gives a near-linear unconditional EF line lower bound. A single PHP_n^{n+1} (n<=12) admitting a tree-like EF derivation with fewer than kappa(n) + n distinct lines refutes the conjecture.

Rationale

Kummer carries control 2-adic structure of binomial coefficients, which in turn govern the counting identities a Frege system must internalize to derive PHP; the carries-matrix rank is a clean, computable surrogate for that structural obstruction. Linking carry-rank to EF line count proposes that the same arithmetic obstruction that makes binomial 2-adic valuations dense also forces many distinct extension formulas. This bridge is computationally tractable (just bit-parity and GF(2) Gaussian elimination) and has, to my knowledge, never been formulated.

Novelty

- Judge: `NOVEL` over 10 arXiv hits

Top hits consulted: - [1309.5828v1] On certain definite integrals and infinite series - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1403.8106v1] Recent advances on the log-rank conjecture in communication complexity - [2412.05017v5] Reduction from the partition problem: Dynamic lot sizing problem with polynomial complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe071b22.py", line 104, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe071b22.py", line 72, in run_trial
    M_n = build_carry_matrix(n)
          ^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe071b22.py", line 35, in build_carry_m
    M[i][j] = add_with_carry(i, j) % 2
              ^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe071b22.py", line 29, in add_with_carr
    return int(''.join(map(str, reversed(result))), 2)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

ValueError: invalid literal for int() with base 2: ''
```

Judge reasoning

The test crashed with a ValueError in build_carry_matrix before any data was produced, so neither the support nor falsification criteria could be evaluated. | next: Fix add_with_carry to handle the empty-result case (e.g., when i=j=0) and re-run the pre-registered protocol over n in {4,6,8,10,12}.

Sharply-Bounded Refutability Vanishes at 3-SAT Threshold

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded arithmetic via the Cook–Reckhow–Krajicek correspondence: Σ^b_0 / PV-equational provability of CNF unsatisfiability, instantiated propositionally as the iterated unit-propagation + depth-1 failed-literal closure operator UPC (the rule fragment that captures polynomial-time PV recursion without nontrivial case analysis, distinct from constant-width resolution and from RUP-style proofs studied in SAT-solving) × Random unsatisfiable 3-CNF formulas at the SAT/UNSAT threshold; specifically the Boolean predicate UPC(F)∈{0,1} indicating whether the closure derives the empty clause without branching
- **Recorded**: 2026-04-27 18:10 UTC
- **Entry ID**: `08881f411f0c`

Statement

Let F be a uniformly random 3-CNF on n=12 variables with m=⌊4.27·n⌋=51 clauses, conditioned on UNSAT (verified by exhaustive DPLL). Define UPC(F)=1 iff the following procedure terminates with ⊥: repeatedly (a) propagate any unit clause, then (b) for each currently unassigned variable x, tentatively assign x=0 and run pure unit-propagation; if conflict, permanently set x=1 (symmetrically for x=1); halt when no progress is possible or ⊥ is derived. Conjecture: Pr[UPC(F)=1] ≤ 0.20

over the conditioned-UNSAT distribution, and the empirical mean over ≥200 UNSAT samples will lie below 0.20 by at least 3 standard errors.

Rationale

The Cook–Reckhow–Krajicek dictionary identifies sharply-bounded Σ^b_0/PV reasoning with rules that lack genuine case analysis—propositionally, exactly unit propagation plus depth-1 failed-literal probing. Random 3-CNFs at the satisfiability threshold are conjectured to encode expanding implication structure that is provably hard for any constant-width or branching-free system, matching the known intuition that PV cannot prove most random combinatorial UNSAT statements. Quantifying the UPC success rate at small n exposes the BA/proof-system gap empirically.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1008.1260v1] Structure of random r-SAT below the pure literal threshold - [2405.16149v3] Small unsatisfiable k-CNFs with bounded literal occurrence - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [1411.7087v6] Consistency proof of a fragment of PV with substitution in bounded arithmetic - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136d91a0.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136d91a0.py", line 69, in run_trial
    cnf = generate_3cnf(n, m)
          ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136d91a0.py", line 22, in generate_3cnf
    if len(clause) == 3 and all(abs(lit) not in clause for lit in cnf):
                            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136d91a0.py", line 22, in <genexpr>
    if len(clause) == 3 and all(abs(lit) not in clause for lit in cnf):
                                ^^^^^^^^
TypeError: bad operand type for abs(): 'tuple'
```

Judge reasoning

The test crashed with a TypeError in generate_3cnf (abs() applied to a tuple) before any trials ran, so no empirical data was produced and the pre-registered support condition cannot be evaluated. | next: Fix the CNF generator: iterate literals from each clause tuple (e.g., `all(abs(lit) not in {abs(l) for l in clause} for cl in cnf for lit in cl)`) and rerun with N≥200 UNSAT-conditioned samples.

Run-Length Entropy Floor Lower-Bounds DNF-MCSP Term Count

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorics on words: run-length encoding (RLE) entropy of the truth-table string, specifically H_RLE(f) = -Σ (l_i/2^n) log2(l_i/2^n) where l_i are the maximal constant-block lengths in the lex-ordered truth table TT(f) ∈ {0,1}$^{\{2}$n} (a Lempel-Ziv-adjacent but distinct

word statistic, rarely used in meta-complexity) × DNF-MCSP: minimum number of product terms DNF_min(f) for a Boolean function f:{0,1}^n→{0,1} given its 2^n-bit truth table; the Hirahara–Oliveira–Santhanam restricted-MCSP proxy for circuit-MCSP

- **Recorded**: 2026-04-27 18:40 UTC
- **Entry ID**: `702dd618723f`

Statement

For every Boolean function f on n≤6 variables with at least one satisfying and one falsifying assignment, DNF_min(f) ≥ ⌈ 2^{H_RLE(f)} / (n+1) ⌉, where H_RLE(f) is the run-length Shannon entropy (in bits) of the lex-ordered truth-table word. Equivalently, exp2(H_RLE(f)) ≤ (n+1)·DNF_min(f) for all such f. A single counterexample — any f with (n+1)·DNF_min(f) < 2^{H_RLE(f)} — refutes the conjecture.

Rationale

DNF_min(f) is bounded by the number of maximal monochromatic subcubes covering supp(f), and each such subcube contributes a contiguous run in TT(f) of length 2^k for some k≤n only along the lowest-indexed coordinate, so RLE block-structure is forced by DNF cover structure; converting the multiplicative lex-ordering bias into Shannon entropy yields a sub-linear-in-2^n floor that is non-trivial for structured f and conjecturally tight up to the (n+1) Boolean-fan-in slack. The bridge — RLE entropy of the truth-table word as an MCSP lower bound — sits in the Hirahara–Oliveira–Santhanam meta-complexity program and does not relativize because it directly inspects the truth-table representation. RLE entropy is computable in O(2^n) time, side-stepping the ‘uncomputable invariant’ failure mode that killed earlier symbolic-dynamics attempts.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1802.00703v2] From Clustering Supersequences to Entropy Minimizing Subsequences for Single and Double Deletions - [1501.01331v1] DNF complexity of complete boolean functions - [1311.0822v1] Properties of maximum Lempel-Ziv complexity strings

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0742e7bb.py", line 121, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0742e7bb.py", line 97, in run_trial
    slack = compute_slack(tt)
            ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0742e7bb.py", line 87, in compute_slack
    h_rle_val = h_rle(tt)
                ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0742e7bb.py", line 53, in h_rle
    entropy = sum(count * math.log2(length + 1) for _, length in rle)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0742e7bb.py", line 53, in <genexpr>
    entropy = sum(count * math.log2(length + 1) for _, length in rle)
                  ^^^^^
```

```
NameError: name 'count' is not defined. Did you mean: 'Counter'?
```

Judge reasoning

The test crashed with a NameError in the h_rle function (undefined variable 'count' in the generator expression) before any trials produced data, so the pre-registered support condition could not be evaluated. | next: Fix the h_rle implementation (e.g., use Counter over run lengths or iterate as 'for length, count in counter.items()') and rerun the ≥800-sample sweep.

Hamming-Weight Spectrum Slope Bounds ACC^0[2] Subcircuit Count

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Krawtchouk-polynomial spectral analysis: the slope at k=n/2 of the weight-distribution polynomial W_f(z)=Σ_{w=0}^{n} A_w(f) z^w of a Boolean function f, where A_w(f)=#{x∈{0,1}^n : |x|=w, f(x)=1}; the slope σ(f):=W_f'(1)/W_f(1) − n/2 is a classical coding-theory invariant (Delsarte/MacWilliams duality) very rarely deployed against circuit lower bounds. × ACC^0[2] circuit complexity for Boolean functions presented by truth-table; specifically the minimum number S_2(f) of {AND,OR,NOT,MOD2}-gates in a depth-3 ACC^0[2] circuit computing f, the Williams/Murray target class for NEXP⊄ACC^0 separations.
- **Recorded**: 2026-04-27 19:11 UTC
- **Entry ID**: `1ad89b4c1108`

Statement

For every Boolean function f:{0,1}^n→{0,1} with n≤20, |σ(f)|·log2(1+S_2(f)) ≥ Var_{x~U}[|x| · f(x)] / (n+1), where Var is taken over the uniform distribution on {0,1}^n. Equivalently: any ACC^0[2] circuit of size s computing f must satisfy s ≥ 2^{Var(|x|·f(x))/((n+1)|σ(f)|)} − 1. A single (f,s) pair with s small but Var/((n+1)|σ(f)|) large refutes the conjecture.

Rationale

MOD2 gates can erase any single Krawtchouk coefficient, but the slope σ(f) is a *signed average* over all weights and is preserved up to a multiplicative log-size factor by depth-3 ACC^0[2] under random restriction (Smolensky-style approximation by low-degree GF(2) polynomials commutes with weight enumeration). The variance of |x|·f captures the layer-wise informational asymmetry that AND/OR gates can collapse only by multiplying fan-in, giving a Krawtchouk-vs-size trade-off in the spirit of Williams' algorithmic-method size bounds.

Novelty

- Judge: `NOVEL` over 12 arXiv hits

Top hits consulted: - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [2407.04826v1] Multi-strategy Based Quantum Cost Reduction of Quantum Boolean Circuits - [2003.09703v1] Variance function of boolean additive convolution - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0054871c.py", line 114, in <module>
    result = run_trial(seed)
```

```
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0054871c.py", line 85, in run_trial
    A_w_values = [A_w(f, w) for w in range(n + 1)]
                  ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0054871c.py", line 30, in A_w
    n = len(f)
        ^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed with a TypeError in A_w (called len() on an int) before any instances were evaluated, so neither the support nor the falsification criterion can be assessed. | next: Fix the A_w helper to accept the truth-table representation actually passed in (or convert f to a sequence/array before calling len), then rerun the full pre-registered sweep.

Pullback monotonicity of coarse depth under coarse-Lipschitz reductions: a functoriality test for κ in CG-KW

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer–Wigderson (CG-KW) framework; underlying math branch: large-scale / coarse geometry of finite metric spaces and Roe C*-algebras (Roe 2003; Nowak–Yu 2012), interfaced with Karchmer–Wigderson communication geometry × Boolean formula depth (NC^1 vs. P) under m-reductions: specifically, the behavior of formula-depth lower bounds along coarse-Lipschitz many-one reductions φ: {0,1}^n → {0,1}^{n'} between explicit functions f ≤_m g (a complexity-theoretic preorder on Boolean functions)
- **Recorded**: 2026-04-27 19:34 UTC
- **Entry ID**: `cc8127953d91`

Statement

Let φ : {0,1}^n → {0,1}^{n'} witness a many-one reduction f ≤*m g, and suppose φ induces a coarse-Lipschitz map Φ : (X_f, d_f) → (X_g, d_g) with distortion D := sup*{x≠y} max(d_g(Φx,Φy)/d_f(x,y), d_f(x,y)/d_g(Φx,Φy)). Then the reduction_pullback Φ^* : C_u^*[X_g] → C_u*^[X_f] and Φ^* : HX^1(X_g) → HX^1(X_f) satisfy the inequality κ(f) ≤ D · κ(g) + O(1). Consequently, the CG-KW target invariant κ is (up-to-distortion) monotone under the m-reduction preorder, so a super-logarithmic κ-lower bound for any g transfers to every f that coarse-Lipschitz reduces to g, yielding an oracle-free, NP-hardness-style closure for κ-based depth lower bounds.

Rationale

This sub-conjecture stress-tests A5 (Roe-index rigidity) together with the categorical glue provided by the reduction_pullback operation. A5 asserts that Roe-trace classes are coarse-equivalence invariants; a reduction with bounded distortion is a quantitative coarse map, so the index pairing ⟨c, T⟩ should pull back with at most a multiplicative D-loss — exactly mirroring how Mendel–Naor metric-cotype invariants and Yu's coarse Baum–Connes invariants behave under quasi-isometries. If the inequality fails experimentally, A5 is false at finite scale; if it holds, κ inherits the m-reduction closure that classical complexity invariants enjoy, which is essential for using CG-KW to lift bounds for one explicit function to many.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [1404.7443v2] Depth Lower Bounds against Circuits with Sparse Orientation - [1411.4413v2] Observation of the rare $B_s^0 \to \mu^+\mu^-$ decay from the combined analysis of CMS and LHCb data - [2507.06261v6] Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capab

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5c3d4406.py", line 187, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5c3d4406.py", line 100, in run_trial
    if f[x].count(1) == 1 and f[y].count(1) == 1:
                              ~^^^
IndexError: tuple index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any trial data, so the pre-registered support condition could not be evaluated. Per the rules, a crashed test yields INCONCLUSIVE. | next: Fix the indexing bug in run_trial (line 100 accesses f[y] out of range — likely a mismatch between tuple length and sampled indices) and rerun the 20-trial protocol.

Ihara Zeta Entropy Lower-Bounds DPLL Depth on 3-Regular Tseitin

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ihara zeta functions of finite graphs (analytic number theory / spectral graph theory in the Stark–Terras–Storm tradition: closed primitive backtrackless cycle counting via the determinant identity ζ_G(u)^{-1} = (1-u$^{2)}${|E|-|V|} · det(I - uA + (d-1)u^2 I)) × DPLL refutation depth (a polynomially-tight proxy for tree-like resolution depth, hence a lower bound on resolution width up to log factors via Ben-Sasson–Wigderson) of Tseitin formulas T(G,c) on small 3-regular graphs G with odd total charge
- **Recorded**: 2026-04-27 19:45 UTC
- **Entry ID**: `3a9492e9db22`

Statement

Define the normalized Ihara entropy of a 3-regular graph G at the regular point u=1/3 by η(G) := (1/|V|) · Σ_i ln|11 - 3·λ_i(A_G)|, where {λ_i} are the adjacency eigenvalues. Then for every connected 3-regular simple graph G on |V|≤20 vertices and every charge c:V→{0,1} with Σ c(v) ≡ 1 (mod 2), the DPLL refutation depth satisfies d_DPLL(T(G,c)) ≥ (1/4) · |V| · η(G), with the inequality strict whenever G is non-bipartite. A single (G,c) violating this bound falsifies the conjecture.

Rationale

The Ihara determinant aggregates spectral information about closed non-backtracking walks, the same combinatorial objects (short cycles in the underlying graph) that drive both expansion and the Ben-Sasson–Wigderson width lower bound for Tseitin; yet ζ_G has essentially never been used in proof complexity (the Terras school treats it as an analytic object). Because expander Tseitin formulas push η(G) toward its 3-regular maximum (eigenvalues concentrated near 0 inflate Σ ln|11-3λ_i|), an Ihara-driven lower bound would refine the BSW expansion bound by capturing higher-order non-backtracking structure invisible to the Cheeger/edge-expansion invariants already explored. The test does not relativize, evades natural proofs (η is highly non-large/non-constructive on random functions because it is a graph spectral statistic, not a function-of-truth-table property), and is algebrization-immune.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1804.08448v8] On Asymptotic Formula for the First Moment of the Riemann Zeta-Function on the Critical Line - [0311029v1] Zeta functions over zeros of general zeta and L-functions - [1302.4644v1] Heat kernels on regular graphs and generalized Ihara zeta function formulas - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [1009.0997v3] Spectral asymptotics for Robin problems with a discontinuous coefficient

Empirical Test

- exit code: `124`, elapsed: `180.10s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode 124) without producing a RESULT line, so no empirical data is available to evaluate the pre-registered support condition. | next: Reduce instance scope (e.g., |V|∈{6,8,10}, ≤10 graphs each) or optimize DPLL depth computation and increase the timeout to obtain measurable data.

Hypercontractive p→2 Norm of Restriction Operator Bounds Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Hypercontractivity / Bonami-Beckner semigroup p→2 norm estimates on the Boolean slice × Resolution refutation width w(F⊢⊥) of small unsatisfiable 3-CNF formulas
- **Recorded**: 2026-04-27 20:12 UTC
- **Entry ID**: `a0e6af1e7a28`

Statement

Let F be an unsatisfiable 3-CNF on n variables, and let A_F:{0,1}^n→{0,1} be the assignment-acceptance indicator (1 iff a partial assignment violates no clause). Define H(F):=sup_{p∈[1,2]} log(||T_{1/sqrt(p-1)} A_F||_2 / ||A_F||_p), the Bonami-Beckner-noise-amplified p→2 hypercontractive defect of A_F evaluated on a finite p-grid {1.1,1.3,...,1.9}. The conjecture: w(F⊢⊥) ≥ ⌈H(F)/log_2(n+1)⌉ for every unsatisfiable 3-CNF F with n≤20, with equality holding for at least 60% of the random unsatisfiable formulas at clause density 4.6.

Rationale

Hypercontractivity controls how mass concentrates on low-degree Fourier levels, and the Ben-Sasson–Wigderson width-degree correspondence relates resolution width to spectral concentration of the unsatisfiable region's indicator. A defect in the Bonami-Beckner inequality at small p quantifies high-degree Fourier energy of A_F, which should obstruct narrow resolution refutations: a narrow refutation forces A_F to be approximated by a low-degree multilinear polynomial. Unlike total-influence approaches, the p→2 norm probes the entire Fourier spectrum at multiple noise scales, capturing graded structure that a single influence number cannot.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1905.12372v1] Resolution Lower Bounds for Refutation Statements - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [0905.1587v5] Unsatisfiable Linear CNF Formulas Are Large and Complex - [0806.1148v2] Unsatisfiable CNF Formulas need many Conflicts

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_30c405ba.py", line 109, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_30c405ba.py", line 83, in run_trial
    if is_satisfiable(clauses):
       ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_30c405ba.py", line 28, in is_satisfiabl
    n = max(abs(c) for clause in clauses)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_30c405ba.py", line 28, in <genexpr>
    n = max(abs(c) for clause in clauses)
                ^
NameError: name 'c' is not defined
```

Judge reasoning

The test crashed with a NameError ('c' is not defined') in the is_satisfiable helper before any trials completed, so neither the violation count nor the equality fraction at density 4.6 could be measured. | next: Fix the generator expression bug (use `max(abs(c) for clause in clauses for c in clause)`) and rerun the pre-registered 640-instance sweep.

Median-Restriction Hardness Amplification Gap for DNF Truth Tables

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Quantile statistics of random subcube restrictions (order-statistic / median-of-medians combinatorics on the Boolean cube; rarely deployed in worst-case-to-average-case amplification, distinct from Hastad/Razborov switching-lemma expectation-style bounds) × DNF-MCSP / minimum DNF term count DNF_min(f) for a Boolean function f:{0,1}^n→{0,1} given its 2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy used in average-to-worst-case meta-complexity reductions
- **Recorded**: 2026-04-27 20:42 UTC
- **Entry ID**: `b62aeb25bb4c`

Statement

Let M_p(f) be the median, over 1024 i.i.d. random p-restrictions ρ (each variable independently *-ed with probability p, else uniform 0/1) with p = 1/sqrt(n), of the term count DNF_min(f|_ρ) of the restricted truth table; let A_p(f) be the corresponding mean. We conjecture that for every Boolean function f on n ≤ 16 variables, DNF_min(f) ≤ 2 · M_p(f)^2 + n, and moreover the median–mean gap satisfies A_p(f) − M_p(f) ≤ sqrt(DNF_min(f)) · log2(n+2). One Boolean function violating either inequality refutes the conjecture.

Rationale

Switching-lemma arguments bound *expected* restricted-DNF size, but average-to-worst-case lifts (Impagliazzo–Wigderson, Hirahara) actually need a tail/quantile statement: a typical restriction must already reveal hardness, not just the expectation. Replacing the mean by the median controls heavy-tailed restriction outliers and conjecturally yields a sharper, square-rooted relation to the unrestricted term count, exposing the structure that lets average DNF-MCSP hardness amplify to worst-case via restriction self-reducibility. The bridge is rare because order statistics of restriction-size functionals (as opposed to expectations) have not, to my knowledge, been used as a complexity invariant.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Total-Influence Defect of Clause-Indicator Bounds Tree-Resolution Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis of Boolean functions: total influence I(g) = sum_i Inf_i(g) and the KKL/Friedgut spectral concentration of the clause-indicator function g_F(x) = AND over clauses C of (clause C is satisfied by x), restricted to its 0-set (the set of UNSAT assignments) — specifically the defect Δ(F) = log2(#UNSAT(F)) − I(g_F)·n/(2·#UNSAT(F)), a low-cost spectral statistic × Tree-like resolution refutation depth d_T(F⊢⊥) of small unsatisfiable 3-CNFs, equivalently DPLL search-tree height on F
- **Recorded**: 2026-04-27 21:18 UTC
- **Entry ID**: `633d44f5ae88`

Statement

For every unsatisfiable 3-CNF F on n≤20 variables with m≤5n clauses, the tree-resolution depth satisfies d_T(F⊢⊥) ≥ ⌈Δ(F)⌉, where Δ(F) := log2(max(1,#UNSAT(F))) − (n · I(g_F)) / (2·max(1,#UNSAT(F))) and I(g_F) = Σ_i Pr_x[g_F(x) ≠ g_F(x⊕e_i)] is computed by direct enumeration of the 2^n cube. Equivalently, the average pivot-influence of g_F lower bounds the DPLL height by a logarithmic-counting correction term.

Rationale

KKL-style total influence quantifies how ‘spread’ the UNSAT region is across coordinate flips; a DPLL tree of depth d partitions {0,1}^n into ≤ 2^d leaves, each a subcube on which g_F is constant, so leaves with many UNSAT points must have small boundary, forcing I(g_F) small unless d is large. The defect Δ(F) makes this trade-off quantitative without invoking switching lemmas, and—unlike entropy-of-leaves arguments already used in proof complexity—it is purely a Walsh-spectral quantity of g_F, an angle largely unexplored for resolution depth.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0209032v3] Complexity Results on DPLL and Resolution - [2003.09703v1] Variance function of boolean additive convolution - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [0806.1148v2] Unsatisfiable CNF Formulas need many Conflicts - [0905.1587v5] Unsatisfiable Linear CNF Formulas Are Large and Complex

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82aca615.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82aca615.py", line 88, in run_trial
    clause = [random.choice([-i-1, i+1] for i in range(n)) for _ in range(3)]
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 346, in choice
    if not len(seq):
           ^^^^^^^^
TypeError: object of type 'generator' has no len()
```

Judge reasoning

The test crashed with a TypeError due to a bug in clause construction (`random.choice` was passed a generator instead of a list), so no instances were ever evaluated and the pre-registered support condition cannot be assessed. | next: Fix the clause sampler (e.g., `random.choice([-i-1, i+1])` for a chosen `i`, or wrap the generator in `list(...)`) and rerun the 80-instance-per-n sweep for n in {8,10,12,14,16}.

Diameter-Multiplicity Power Law under Controlled Refinement Recovers Coarse Depth

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); coarse geometry / Roe-style coarse cohomology of finite KW metric spaces (Gromov asymptotic dimension, Roe algebras) × Formula-depth complexity (NC^1 vs. P): obtaining super-logarithmic formula-depth lower bounds for explicit f by reading off the multiplicity exponent of a controlled cover of X_f
- **Recorded**: 2026-04-27 21:34 UTC
- **Entry ID**: `ed6023df7533`

Statement

Let f:{0,1}^n -> {0,1} be explicit and let X_f be its KW-metric space with metric d_f. For any sequence of controlled covers U_0, U_1, U_2, ... of X_f produced by iterated controlled_refinement, with diameters R_k = R_0 · 2^{-k} and multiplicities m_k, the conjecture is that the asymptotic exponent γ(f) := limsup_{k→∞} log(m_k) / log(R_0/R_k) satisfies γ(f) = κ(f) up to additive O(1). Equivalently: the slope of log(m_k) versus log(R_0/R_k) recovers the coarse depth invariant from the framework's target, and hence by axiom A1 lower-bounds depth(f). In particular for an explicit family with γ(f_n) = ω(log n), one obtains depth(f_n) = ω(log n).

Rationale

This sub-conjecture directly tests the proof mechanism asserted by axiom A1, which states that the multiplicity exponent of a controlled cover equals depth (the cover is built from a KW protocol tree, and refinement halves diameter at the cost of one extra fan-out at each gate, so multiplicity should compound at the depth rate). It also exercises the controlled_refinement operation as a numerical probe rather than a structural one, providing an alternative empirical handle on κ(f) that bypasses construction of explicit cocycles in HX^1. If A1's proof is correct, the diameter–multiplicity log-log slope must match κ(f) on independently checkable small instances; deviation would signal a gap in A1's translation step. It is independent of the previously listed conjectures because it concerns the refinement operation rather than the trace pairing, the ultralimit, oracle collapse, pullbacks, or composition.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Commutator-Length Defect in S_5 Predicts Width-5 BP Size of Boolean Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial group theory of S_5: commutator length cl(g) (minimum k with g = [a_1,b_1]…[a_k,b_k]) and its slack against the trivial bound, computed by exact BFS in the Cayley graph of S_5 under conjugation moves × Width-5 permutation branching program (BP) length L_5(F) representing a Boolean formula F via Barrington's construction; equivalently a quantitative formula-size proxy in NC^1
- **Recorded**: 2026-04-27 21:48 UTC
- **Entry ID**: `bbfdc40acf2e`

Statement

For every Boolean formula F over n<=20 variables and depth d<=6, let phi(F) be the unique 5-cycle product produced by Barrington's standard inductive construction on F, and define the commutator-length defect D(F) = 4*depth(F) - cl(phi(F)) where cl is the S_5 commutator length of phi(F) when 1=accept, 1=reject distinguishes by 5-cycles. Then the optimal width-5 BP length satisfies L_5(F) >= 2^{D(F)}* (2*depth(F) + 1), i.e., a positive commutator-length defect forces an exponential-in-defect blow-up over the trivial 4^depth Barrington bound. Concretely: every formula F with D(F) >= 1 admits no width-5 BP of length < 2*(2*depth(F)+1).*

Rationale

Barrington's theorem encodes formula evaluation by iterated commutators of 5-cycles, but the only structural invariant ever extracted from the resulting permutation is its cycle type; commutator length in S_5 is a finer, rarely-used algebraic statistic that measures how ‘cheaply’ phi(F) can be re-expressed as commutators, potentially detecting hidden algebraic redundancy that a shorter

BP would have to exploit. If a short BP exists, the realized permutation must be writable with few commutators, so a large defect should obstruct compression; this connects geometric group theory's stable commutator length tradition (Calegari, Bavard) to NC^1 lower bounds in a way that has, to our knowledge, not been pursued.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fa494294.py", line 117, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fa494294.py", line 84, in run_trial
    phi_F = Barrington_construction(F)  # Assume this function is defined elsewhere
            ^^^^^^^^^^^^^^^^^^^^^^^
NameError: name 'Barrington_construction' is not defined
```

Judge reasoning

The test crashed with a NameError because Barrington_construction was never defined, so no data was produced and the pre-registered support condition cannot be evaluated. | next: Implement Barrington's standard inductive construction (mapping AND/OR/NOT gates to 5-cycle products in S_5) and re-run the trial to actually compute D(F) and L_5(F).

Hessian-Rank of Clause-Count Polynomial Bounds DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory: the Mignon-Ressayre Hessian-rank lower bound for determinantal complexity dc(p) — rank of Hess(p) at a smooth point lower-bounds 2·dc(p); the most computationally tractable orbit-closure obstruction in the Mulmuley-Sohoni / Bürgisser-Ikenmeyer lineage. × DPLL search-tree depth d_DPLL(F) of unsatisfiable 3-CNF formulas F at clause density α=5.0 (lex-DPLL with unit propagation), equivalently tree-resolution depth via the standard correspondence.
- **Recorded**: 2026-04-27 22:23 UTC
- **Entry ID**: `67153fba470b`

Statement

For every UNSAT 3-CNF F on n ∈ [6,11] variables at clause density α=5.0, define p_F(x) := Σ_{C ∈ F} Π_{l ∈ C}(1 - l*) ∈ Q[x_1,...,x_n], the degree-3 multilinear 'unsatisfied-clause counter' (l* := x_i for positive literal x_i, l* := 1-x_i for negative). Let H_F(x) be its n×n symmetric Hessian matrix of linear forms and r(F) := max_{x ∈ {0,1,2}^n} rank_Q(H_F(x)). Then r(F) ≥ d_DPLL(F); a single F with r(F) < d_DPLL(F) refutes the conjecture.

Rationale

GCT lower bounds for determinantal/permanental orbit closures classically route through Hessian rank at a smooth point (Mignon-Ressayre 2004; Landsberg-Manivel-Ressayre), and that invariant is the canonical computable GCT witness. The clause-count polynomial p_F is the natural degree-3

GL_n-equivariant arithmetization of an UNSAT 3-CNF: its Hessian at integer evaluation points is a small integer matrix whose rank measures the 'algebraic spread' of clause-variable-pair couplings in F. Linking Mignon-Ressayre rank to DPLL depth would import GCT-style algebraic-geometry obstructions into proof complexity — a bridge essentially unexplored — while staying within concretely computable invariants.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0209032v3] Complexity Results on DPLL and Resolution - [1504.00151v1] Bi-polynomial rank and determinantal complexity - [1403.8106v1] Recent advances on the log-rank conjecture in communication complexity - [1002.1290v3] Bounds on Threshold of Regular Random k-SAT - [0906.0693v3] An improved lower bound on the counterfeit coins problem

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1a6b4ba.py", line 104, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1a6b4ba.py", line 72, in run_trial
    d_DPLL_F = lex_dpll(F, [None] * n)
               ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1a6b4ba.py", line 40, in lex_dpll
    if all(new_assignment[v] == (F[i][v] > 0) * 2 - 1 for i in range(len(F))):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1a6b4ba.py", line 40, in <genexpr>
    if all(new_assignment[v] == (F[i][v] > 0) * 2 - 1 for i in range(len(F))):
                          ^
NameError: name 'v' is not defined
```

Judge reasoning

The test crashed with a NameError ('v' is not defined) in lex_dpll before any instances were evaluated, so no data was produced to assess the support or falsification condition. | next: Fix the scoping bug in lex_dpll (the genexpr references undefined 'v'; likely should iterate over variables in the clause) and rerun the trial.

Herbrand-Disjunction Length of PHP Bounds Tree-Resolution Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded arithmetic via Herbrand's theorem in the Cook-Reckhow-Krajicek correspondence: the Herbrand-disjunction length H_k(F) of a refutable bounded formula F, computed as the minimum number of ground term-tuples t_1,…,t_m needed so that the disjunction /_i ¬φ(t_i) (with φ the existential matrix of F's Skolemization) is a propositional tautology under the natural CNF coding — a PV/S^1_2 invariant that, by Buss's witnessing theorem, polynomially relates to PV-proof length but is rarely measured directly on small instances × Tree-like resolution leaf count L_T(PHP_n^{n+1} ⊢ ⊥) of the pigeonhole principle CNF on n+1 pigeons and n holes, equivalently DPLL search-tree leaves on the standard PHP encoding
- **Recorded**: 2026-04-27 22:52 UTC
- **Entry ID**: `9152edf3c7fc`

Statement

For every n with 3 ≤ n ≤ 10, the Herbrand-disjunction length H(PHP_n^{n+1}) — defined operationally as the minimum number m of pigeon-hole assignment tuples (a_1,b_1),...,(a_m,b_m) such that the propositional disjunction /*{i≤m} (x*{a_i,b_i} ∧ x_{a_i,b'_i}) with b'_i ≠ b_i covers every total injection candidate witnessed by Buss-style elimination — satisfies log_2(L_T(PHP_n^{n+1})) = (1 ± 0.25) · log_2(H(PHP_n^{n+1})) + o(n). One pair (n, witnessed mismatch outside the ±0.25 window with both quantities exactly computed) refutes the conjecture.

Rationale

Cook-Reckhow-Krajicek tie PV/S^1_2 provability of UNSAT to short tree-Frege/resolution refutations via Herbrand witnessing, but no one has empirically measured the Herbrand-witness count side-by-side with DPLL leaf count on PHP, the canonical hard instance. If the bounded-arithmetic correspondence is tight at small scale, the two logs should track linearly; deviation would localize where Buss's witnessing slack hides. PHP avoids relativization (oracle-uniform) and is the textbook target separation between PV-style and stronger theories.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1608.07792v1] A Generalized Resolution Proof Schema and the Pigeonhole Principle - [1411.3304v1] On the complexity of finding falsifying assignments for Herbrand disjunctions - [0304041v1] $P \neq NP$, propositional proof complexity, and resolution lower bounds for the weak pigeonhole principle - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [0906.0693v3] An improved lower bound on the counterfeit coins problem

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 88, in run_trial
    leaves = lex_dpll(cnf, {})
             ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 42, in lex_dpll
    return dfs()
           ^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 38, in dfs
    if dfs():
       ^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 38, in dfs
    if dfs():
       ^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 38, in dfs
    if dfs():
       ^^^^^
  [Previous line repeated 993 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06300a1e.py", line 21, in dfs
    var = next((v for v in range(len(cnf)) if v not in assignment), None)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

The test crashed with a RecursionError in lex_dpll before producing any RESULT line, so neither L_T nor H was computed for any n. Per Rule 2, a crashed test yields INCONCLUSIVE. | next: Rewrite lex_dpll iteratively (or raise sys.setrecursionlimit and fix the variable-selection bug iterating over range(len(cnf)) instead of variable indices), then re-run for n=3..10 across the 5 fixed seeds.

Beck-Fiala Slack of Clause-Variable Hypergraph Bounds DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy theory: the Beck-Fiala slack delta(H) = max_S |sum_{e in S} chi(e)| - 2t+1 of a t-uniform set system H, computed via the Beck-Fiala floating-color rounding algorithm (iteratively freeze tight constraints; rarely deployed against proof complexity) × DPLL search-tree leaf count L_DPLL(F) of small unsatisfiable 3-CNF formulas under a fixed lex variable order with unit propagation, equivalently tree-resolution leaf count
- **Recorded**: 2026-04-27 23:24 UTC
- **Entry ID**: `dbaf5e8ccbe1`

Statement

Let F be an unsatisfiable 3-CNF on n<=20 variables with m clauses, and let H_F be the 3-uniform hypergraph whose vertices are the m clauses and whose hyperedges are the variable-stars E_v = {C : v in C}; let delta(F) be the Beck-Fiala slack of H_F obtained by Beck-Fiala floating-color rounding (max over all variable subsets S of |sum_{v in S} chi(E_v)| minus the trivial 2t-1=5 bound, where chi:clauses->{-1,+1} is the discrepancy coloring). Then for every unsatisfiable 3-CNF F at clause density alpha in [4.0,5.5], log2(L_DPLL(F)) >= delta(F)/3 - 1. A single F with delta(F)/3 - 1 > log2(L_DPLL(F)) + 0.5 falsifies the conjecture.

Rationale

Beck-Fiala discrepancy of the variable-star hypergraph quantifies the unavoidable imbalance any +/-1 coloring of clauses must accept, which intuitively obstructs tree-resolution from finding a small witnessing subtree because each branch must resolve a near-balanced set of clauses; this connects classical Beck-Fiala slack (a Chazelle/Matousek-style dispersion invariant) to DPLL leaf complexity through clause coloring rather than variable coloring. Unlike subdeterminant or Kashin-split approaches that target rigidity, this targets proof complexity directly via a hypergraph attached canonically to F. The mapping uses only Beck-Fiala floating-color rounding (~30 lines of pure Python) and standard lex-DPLL, sidestepping advanced algebraic machinery and natural-proofs concerns since delta(F) is not a property of a Boolean function class but of a specific CNF.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2602.09948v1] Non-Additive Discrepancy: Coverage Functions in a Beck-Fiala Setting - [2508.01937v1] An Improved Bound for the Beck-Fiala Conjecture - [1306.6081v3] An improvement of the Beck-Fiala theorem - [0905.1587v5] Unsatisfiable Linear CNF Formulas Are Large and Complex - [0806.1148v2] Unsatisfiable CNF Formulas need many Conflicts

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41bdba6b.py", line 119, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41bdba6b.py", line 88, in run_trial
    m = int(n * alpha)
                ^^^^^
NameError: name 'alpha' is not defined
```

Judge reasoning

The test crashed with a NameError ('alpha' is not defined) before producing any data, so no instances were evaluated against the pre-registered criterion. | next: Fix the scoping bug in run_trial so that alpha is passed in or defined in scope, then rerun the 200-instance sweep.

Monotone Ultralimit Convergence of κ along the Asymptotic Completion of the Majority Family

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse geometry and Roe-algebraic K-theory within combinatorial topology × Formula-depth complexity in the De Morgan basis (NC^1 vs. P lower bounds for Majority/threshold), specifically the logarithmic depth of MAJ_n and its threshold cousins
- **Recorded**: 2026-04-27 23:36 UTC
- **Entry ID**: `2ee2e2f64a25`

Statement

Let {(X_{MAJ_n}, d_{MAJ_n})} be the KW-metric spaces of the Majority family for odd n. Let c_H ∈ HX^1(X_{MAJ_n}) be the canonical Hamming cocycle c_H(x,y) = d_{MAJ_n}(x,y) mod 2, and for each propagation R let T_R ∈ C_u^*[X_{MAJ_n}] be the indicator-banded Roe operator T_R(x,y) = 1[d_{MAJ_n}(x,y) ≤ R]. Then (i) the trace-pairing-derived invariant κ_n := sup_R log|⟨c_H, T_R⟩| / log(R) is monotone non-decreasing in n, and (ii) κ_n ≥ α · log n for some absolute α > 0, so that under asymptotic_completion the limit κ_∞ = lim κ_n is unbounded. Combined with axiom A1 (Coarse-KW link), this predicts depth(MAJ_n) = Ω(log n), and combined with A5 (Roe-index rigidity) the witness is stable under coarse equivalences induced by reduction_pullback.

Rationale

This sub-conjecture stresses two axioms that prior sub-conjectures left untested in tandem: A1 (the Coarse-KW link, in its asymptotic, 'completion' formulation) and the convergence behavior implicit in the asymptotic_completion operation. If κ behaves like a true coarse invariant of an ultralimit, then along any natural function family it should be monotone and divergent whenever depth grows; Majority — whose formula depth is Θ(log n) — is the smallest non-trivial witness where the predicted κ-growth is detectable on tiny instances, but still nontrivial. The Hamming cocycle c_H is the simplest nontrivial coarse 1-cocycle (its δ-image is supported on triangle-defects, which vanish modulo 2 on the bipartite f^{-1}(0) ⊔ f^{-1}(1) partition), and indicator-banded T_R is the canonical Roe-controlled operator; the trace pairing reduces to a counting identity that scales with the propagation R, exactly the quantity that A1 forces to be polynomial in 2^depth. Failure of monotonicity or sub-logarithmic growth would falsify A1 in its strong form; success gives small-scale empirical evidence for the conjectured ultralimit convergence underlying A5.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Spectral-Tail Mass of Unit-Propagation Indicator Bounds DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis of Boolean functions: high-degree Walsh spectrum tail mass W^{>k}[g] = sum_{|S|>k} hat g(S)^2 of the unit-propagation success indicator × DPLL search-tree leaf count L(F) on small unsatisfiable 3-CNF formulas under a fixed lex variable order with unit propagation
- **Recorded**: 2026-04-27 23:55 UTC
- **Entry ID**: `9edcbf300e22`

Statement

For an unsatisfiable 3-CNF F on n variables with m clauses, define g_F:{-1,+1}^n -> {0,1} by g_F(x)=1 iff iterated unit propagation on F under the partial assignment x reaches the empty clause. Let T_k(F) = sum_{|S|>k} hat g_F(S)^2 be the degree->k Walsh tail. Then for every unsatisfiable 3-CNF with n<=20 and m<=5n, log2(L(F)+1) <= 4 * (1 + T_{log2(m+1)}(F) * 2^n) / max(1, hat g_F(emptyset) * 2^n), and for at least 60% of random unsatisfiable instances at clause density 4.5 the ratio (log2(L(F)+1)) / (1 + T_{log2(m+1)}(F) * 2^n / max(1, hat g_F(emptyset) * 2^n)) lies in [0.05, 4].

Rationale

Unit propagation is a low-degree Boolean process: each propagation step depends on at most 3 literals, so the indicator g_F should have most of its Fourier mass on low-degree coefficients when F is locally easy, and DPLL backtracks exactly where g_F's high-degree tail concentrates (KKL/Friedgut intuition: rare, dispersed sensitivities force junta failure). Quantifying the high-degree mass of g_F therefore offers a hypercontractivity-flavored explanation for DPLL leaf blow-up that, unlike total influence on the SAT indicator, is computable directly from F by simulating UP on every assignment.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients - [2406.18700v4] Structure of sparse Boolean functions over Abelian groups, and its application to testing - [2205.09222v2] On the Walsh and Fourier-Hadamard Supports of Boolean Functions From a Quantum Viewpoint - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [1402.4455v1] Symbiosis of Search and Heuristics for Random 3-SAT

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e919bf9.py", line 129, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e919bf9.py", line 61, in run_trial
    if is_unsat(F):
       ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e919bf9.py", line 36, in is_unsat
    propagate()
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e919bf9.py", line 27, in propagate
    pos = any(l > 0 and assignment[l] == 1 for l in clause)
                                                    ^^^^^^
NameError: cannot access free variable 'clause' where it is not associated with a value in enclosing sc
```

Judge reasoning

The test crashed with a NameError in the propagate() function before producing any data, so no support or falsification evidence was generated. Per rule 2, a crash yields INCONCLUSIVE. | next: Fix the scoping bug in propagate() (capture `clause` properly in the generator expression, e.g., bind it as a loop variable or pass it explicitly) and rerun the test.